



UNIVERSITÄT
LEIPZIG

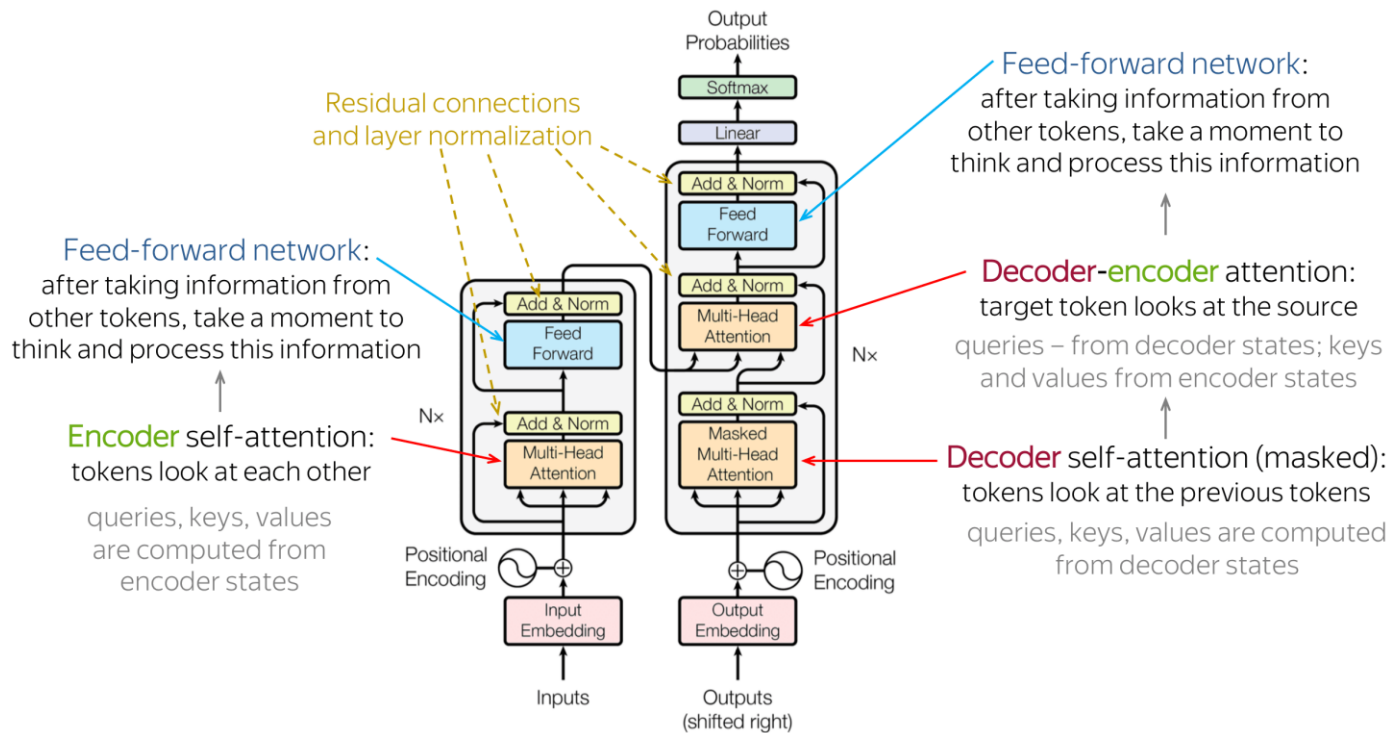
LLMs 2: GPT

Natural Language Processing

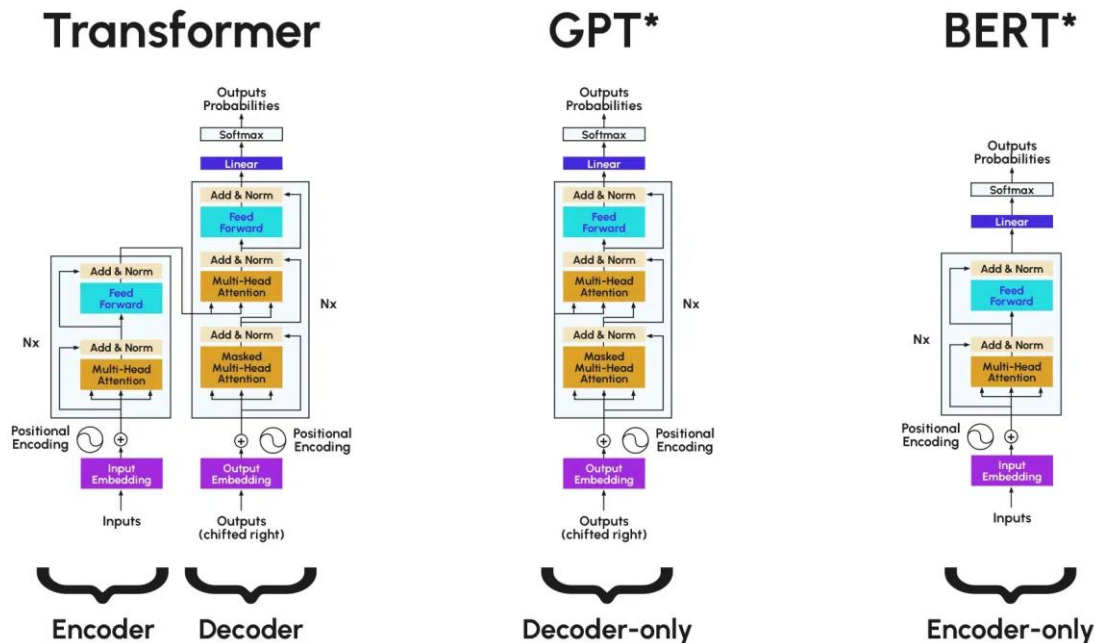
Leonie Weissweiler

RECAP

PUTTING IT ALL TOGETHER



PREVIEW: ENCODER-ONLY AND DECODER-ONLY

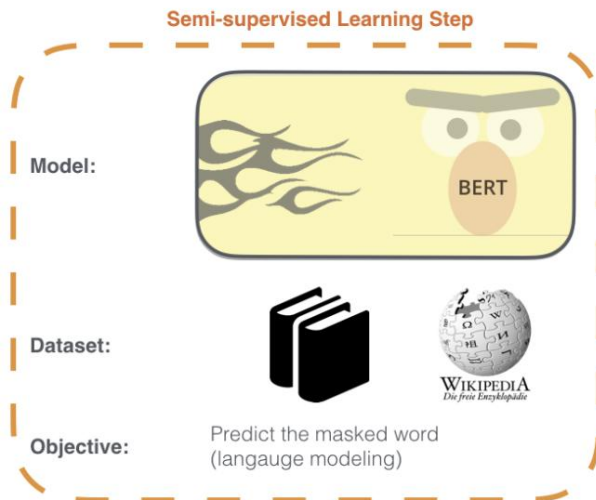


*Illustrative example, exact model architecture may vary slightly

GPT

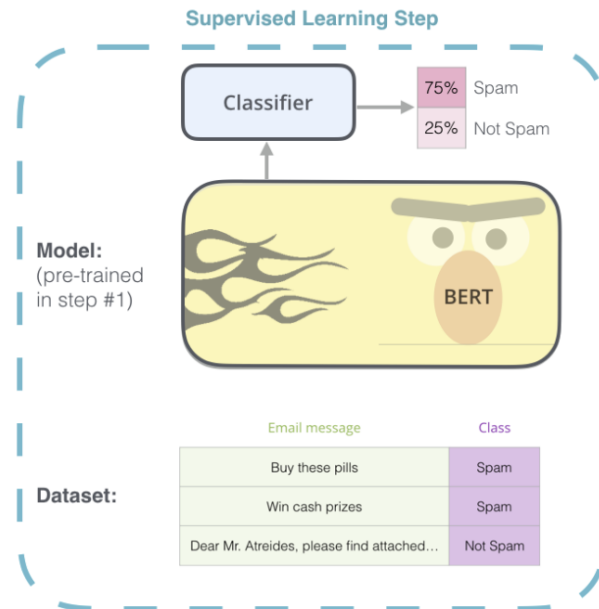
1 - **Semi-supervised** training on large amounts of text (books, wikipedia..etc).

The model is trained on a certain task that enables it to grasp patterns in language. By the end of the training process, BERT has language-processing abilities capable of empowering many models we later need to build and train in a supervised way.



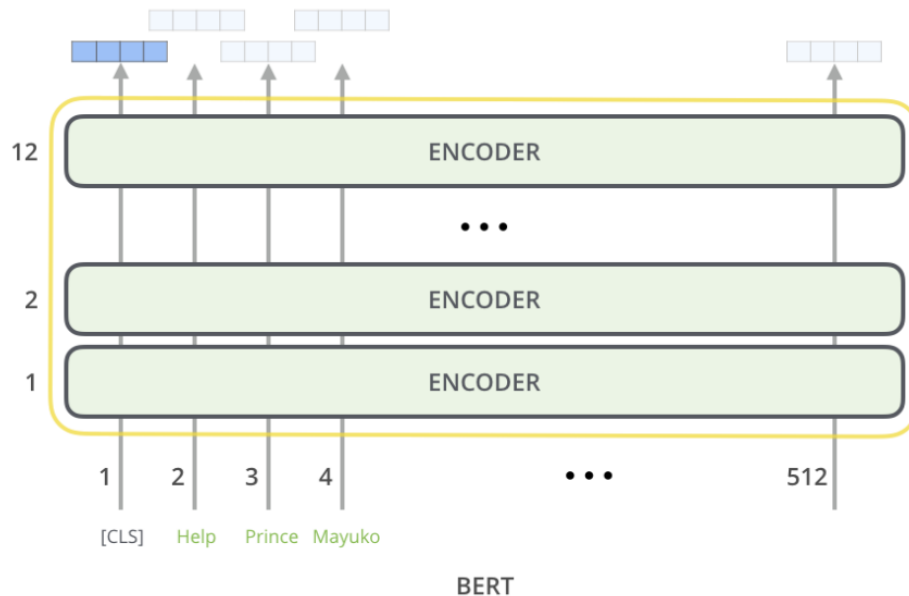
Pretraining: train the language model to predict the masked word, on large amounts of unlabelled text

2 - **Supervised** training on a specific task with a labeled dataset.



Finetuning: put a classification head (essentially a small FFN with an output layer of appropriate size) on top of the pretrained model, then finetune the entire thing with the labelled data for a specific task

BERT OUTPUT



One vector of hidden size (e.g. 768 for BERT Base) for each input token

TWO PRETRAINING OBJECTIVES: MLM AND NSP

Training time: predict if sentences
are consecutive (NSP objective)

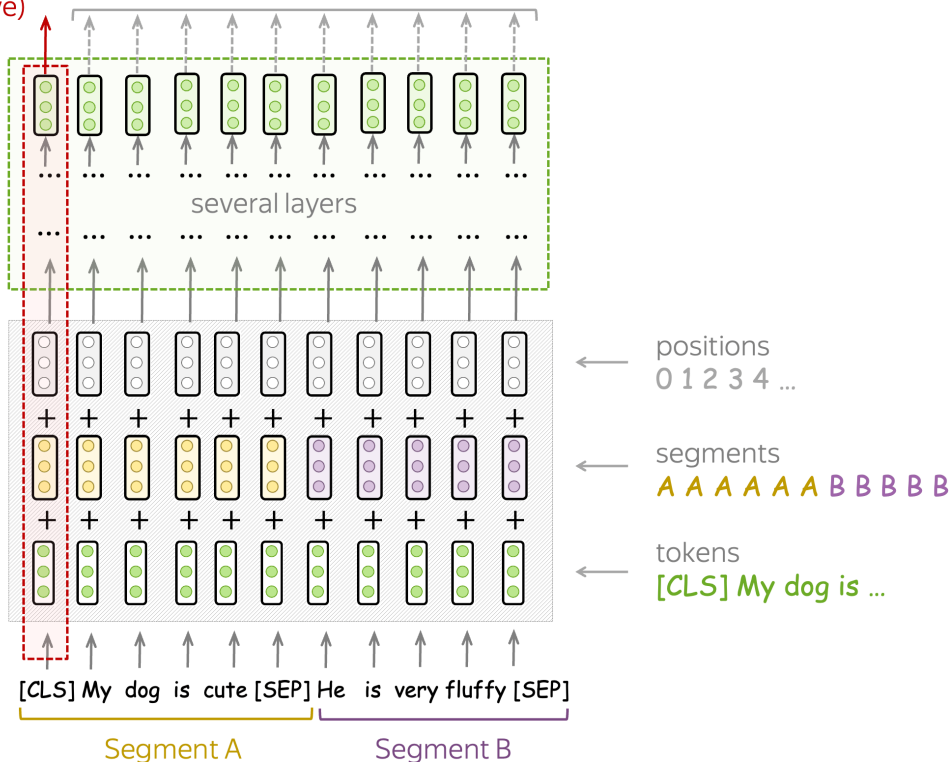
Test time: classification

Training time: MLM objective

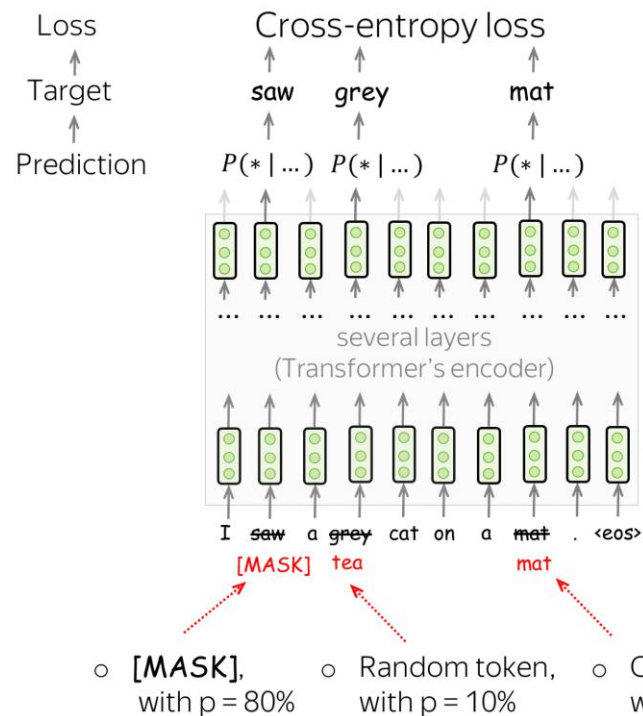
Model
(Transformer
encoder)

Input

Training on pairs of
sentences: either
consecutive or
random (50%/50%)



MASKED LANGUAGE MODELLING



At each training step:

- pick randomly 15% of tokens
- replace each of the chosen tokens with something
- predict original chosen tokens

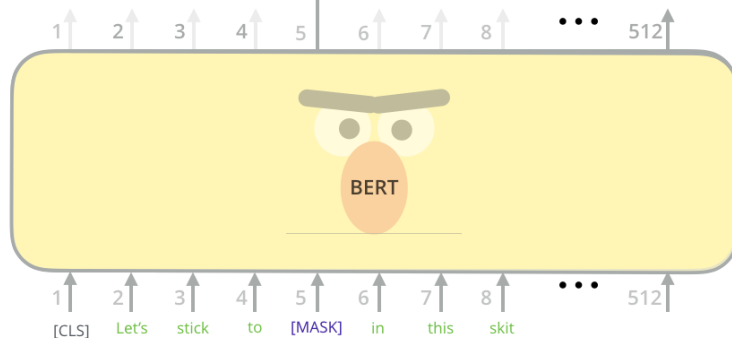
MASKED LANGUAGE MODELLING

Use the output of the masked word's position to predict the masked word

Possible classes:
All English words

0.1%	Aardvark
...	...
10%	Improvisation
...	...
0%	Zyzyva

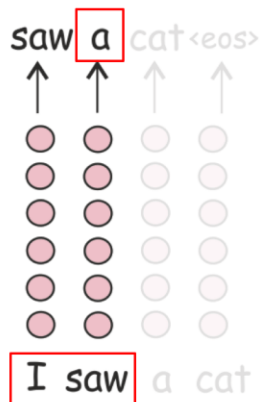
FFNN + Softmax



MLM VS. AUTOREGRESSIVE LANGUAGE MODELLING

Language Modeling

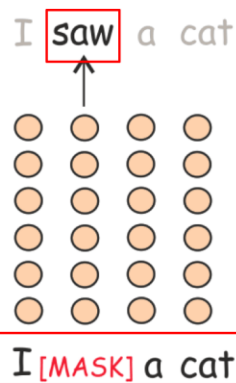
- Target: next token
- Prediction: $P(* | \text{I saw})$



left-to-right, does
not see future

Masked Language Modeling

- Target: current token (the true one)
- Prediction: $P(* | \text{I [MASK] a cat})$



sees the whole text, but
something is corrupted

1-NEAREST NEIGHBOUR ALGORITHM FOR WSD

Training

- Take a sense-labelled corpus, e.g. SEMCOR
- Run the corpus through BERT to get contextual embeddings for each token
- Then for each sense s of a word w for n tokens of that sense, pool the embeddings:

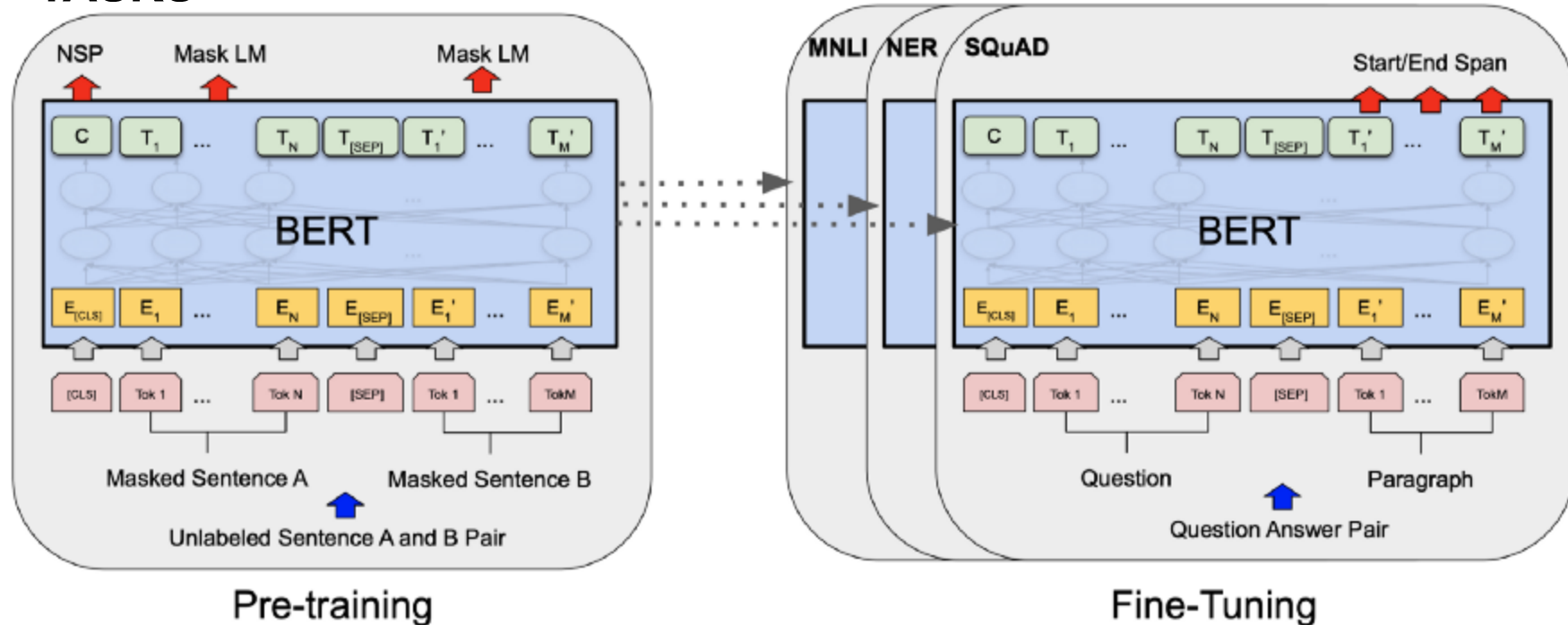
$$\mathbf{v}_s = \frac{1}{n} \sum_i \mathbf{v}_i \quad \forall \mathbf{v}_i \in \text{tokens}(s)$$

Testing

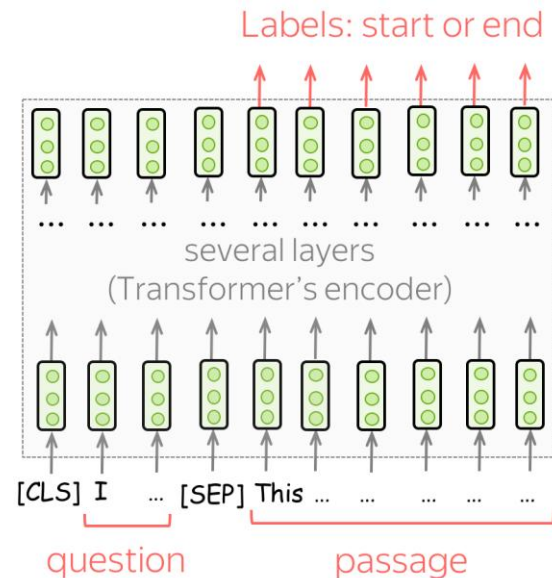
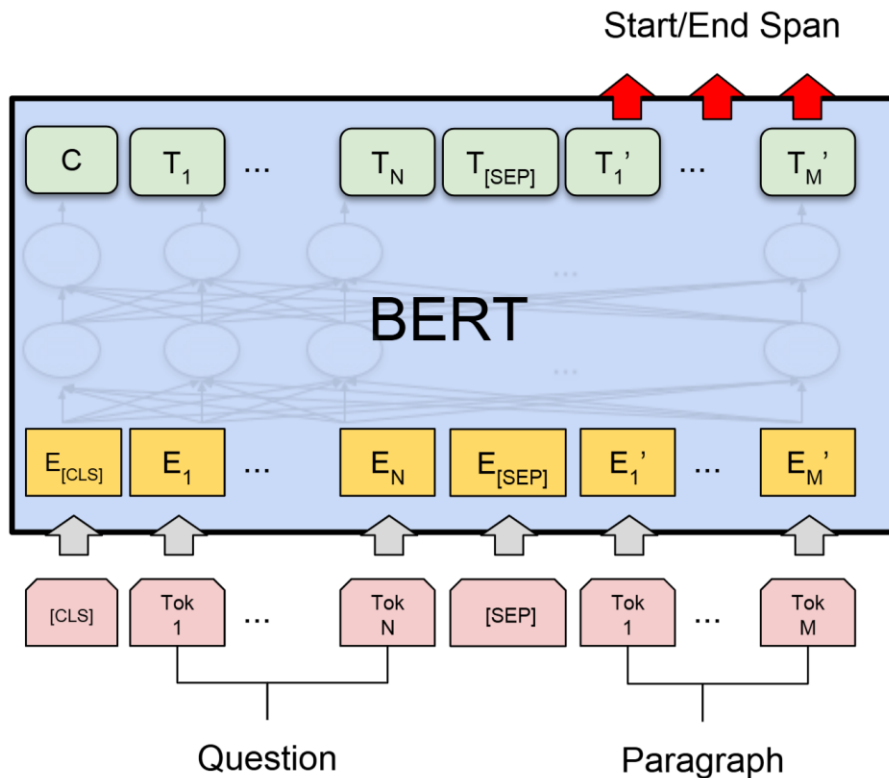
Given a token of a target word t , compute the contextual embedding of t and choose its nearest neighbour sense from the training set

$$\text{sense}(t) = \underset{s \in \text{senses}(t)}{\text{argmax}} \text{cosine}(\mathbf{t}, \mathbf{v}_s)$$

TRANSFER LEARNING: FINE-TUNING BERT ON DIFFERENT TASKS



QUESTION ANSWERING WITH BERT

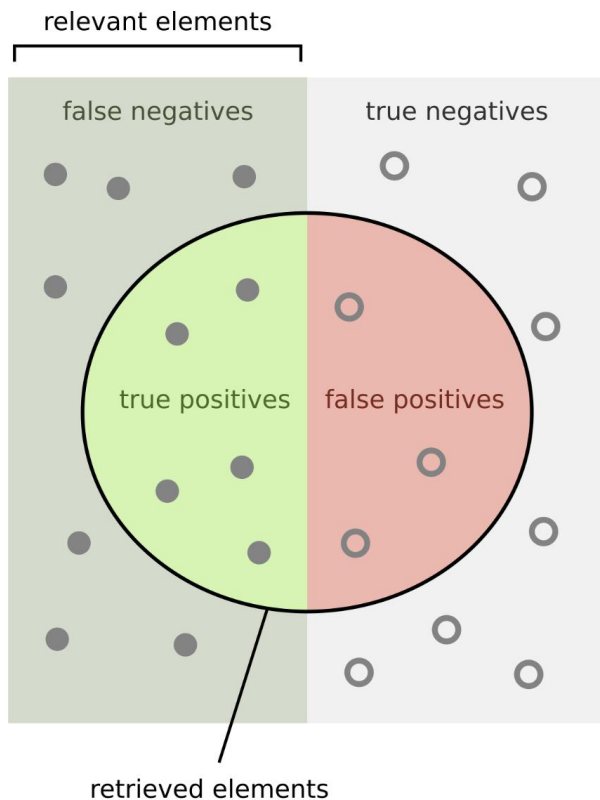


For each token in the text passage where the answer could be, use the final representations to predict if this token is the start or the end of the correct segment

ROBERTA

- RoBERTa: Robustly optimised BERT approach
- The best kind of empirical work: re-implemented BERT, tried out which of the components were useful, found improvements for some
- We can find explanations for why each improvement works, but it was really just empirically determined and all explanations are post-hoc
- Built a better model that is now the standard for MLM

PRECISION AND RECALL



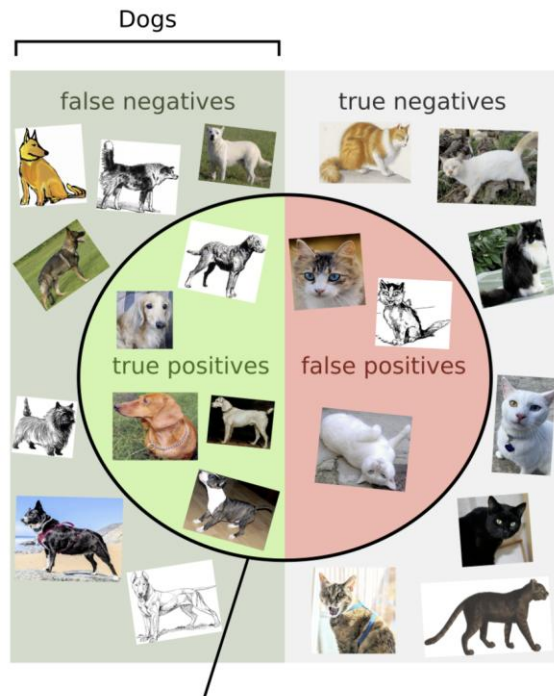
How many retrieved items are relevant?

$$\text{Precision} = \frac{\text{true positives}}{\text{true positives} + \text{false positives}}$$

How many relevant items are retrieved?

$$\text{Recall} = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}}$$

PRECISION AND RECALL - EXAMPLE



images classified as having dogs

$$\text{Precision} = \frac{5 \text{ true pos.}}{8 \text{ total pos.}}$$

$$\text{Recall} = \frac{5 \text{ true pos.}}{12 \text{ total dogs}}$$

$$\text{Accuracy} = \frac{5 \text{ true pos.} + 7 \text{ true neg.}}{22 \text{ total images}}$$

CONFUSION MATRIX

		Predicted condition	
		Predicted positive	Predicted negative
Actual condition	Total population = P + N		
	Real Positive (P) ^[a]	True positive (TP), hit ^[b]	False negative (FN), miss, underestimation
	Real Negative (N) ^[d]	False positive (FP), false alarm, overestimation	True negative (TN), correct rejection ^[e]

Useful to print out when doing classification, to see what the specific problems are

F1-SCORE

$$F_1 = \frac{2}{\text{recall}^{-1} + \text{precision}^{-1}} = 2 \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}} = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}}$$

- F1-Score is the harmonic mean between precision and recall
- Called F1-score because it's the special case of $\beta=1$ of the F-score:

$$F_\beta = (1 + \beta^2) \cdot \frac{\text{precision} \cdot \text{recall}}{\beta^2 \cdot \text{precision} + \text{recall}}$$

But other values of β are used very rarely in ML

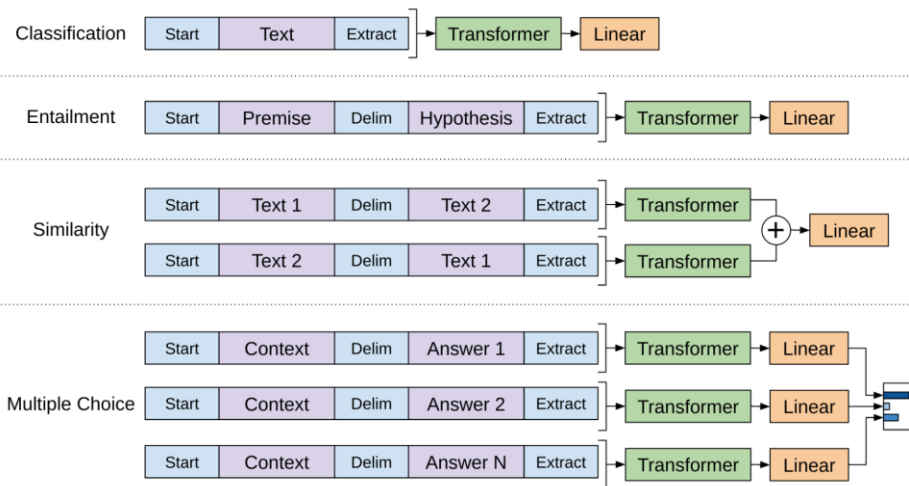
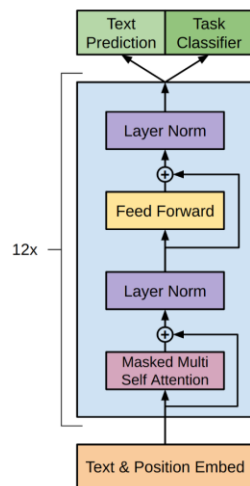
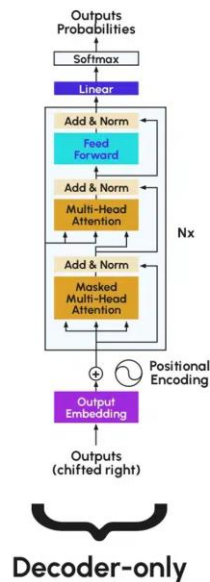
QUESTIONS?

DECODER-ONLY MODELS

GENERATIVE PRE-TRAINED TRANSFORMER (GPT)

GPT*

Everything is conditional generation now



GPT2: NO MORE FINETUNING

- New, larger dataset collected by filtering CommonCrawl: WebText (40GB).
- Bigger models:
- More data but fewer epochs:

Parameters	Layers	d_{model}
117M	12	768
345M	24	1024
762M	36	1280
1542M	48	1600

Dataset	Quantity (tokens)	Weight in training mix	Epochs elapsed when training for 300B tokens
Common Crawl (filtered)	410 billion	60%	0.44
WebText2	19 billion	22%	2.9
Books1	12 billion	8%	1.9
Books2	55 billion	8%	0.43
Wikipedia	3 billion	3%	3.4

THE MAGIC INGREDIENT: NATURALLY OCCURRING DEMONSTRATIONS IN THE TRAINING DATA

- GPT2 wasn't finetuned on any task and still succeeded at all of them
- The likely causes
 - Size (magic?)
 - Training data sometimes resembles tasks

"I'm not the cleverest man in the world, but like they say in French: **Je ne suis pas un imbecile** [I'm not a fool].

In a now-deleted post from Aug. 16, Soheil Eid, Tory candidate in the riding of Joliette, wrote in French: "**Mentez mentez, il en restera toujours quelque chose**," which translates as, "Lie lie and something will always remain."

"I hate the word 'perfume,'" Burr says. 'It's somewhat better in French: 'parfum.'

If listened carefully at 29:55, a conversation can be heard between two guys in French: "**-Comment on fait pour aller de l'autre côté? -Quel autre côté?**", which means "**- How do you get to the other side? - What side?**".

If this sounds like a bit of a stretch, consider this question in French: **As-tu aller au cinéma?**, or **Did you go to the movies?**, which literally translates as Have-you to go to movies/theater?

"Brevet Sans Garantie Du Gouvernement", translated to English: **"Patented without government warranty"**.

Table 1. Examples of naturally occurring demonstrations of English to French and French to English translation found throughout the WebText training set.

THE RESULT: GPT2 CAN ANSWER MANY QUESTIONS THAT WEREN'T IN THE TRAINING DATA

Question	Generated Answer	Correct	Probability
Who wrote the book the origin of species?	Charles Darwin	✓	83.4%
Who is the founder of the ubuntu project?	Mark Shuttleworth	✓	82.0%
Who is the quarterback for the green bay packers?	Aaron Rodgers	✓	81.1%
Panda is a national animal of which country?	China	✓	76.8%
Who came up with the theory of relativity?	Albert Einstein	✓	76.4%
When was the first star wars film released?	1977	✓	71.4%
What is the most common blood type in sweden?	A	✗	70.6%
Who is regarded as the founder of psychoanalysis?	Sigmund Freud	✓	69.3%
Who took the first steps on the moon in 1969?	Neil Armstrong	✓	66.8%
Who is the largest supermarket chain in the uk?	Tesco	✓	65.3%
What is the meaning of shalom in english?	peace	✓	64.0%
Who was the author of the art of war?	Sun Tzu	✓	59.6%
Largest state in the us by land mass?	California	✗	59.2%
Green algae is an example of which type of reproduction?	parthenogenesis	✗	56.5%
Vikram samvat calender is official in which country?	India	✓	55.6%
Who is mostly responsible for writing the declaration of independence?	Thomas Jefferson	✓	53.3%
What us state forms the western boundary of montana?	Montana	✗	52.3%
Who plays ser davos in game of thrones?	Peter Dinklage	✗	52.1%
Who appoints the chair of the federal reserve system?	Janet Yellen	✗	51.5%
State the process that divides one nucleus into two genetically identical nuclei?	mitosis	✓	50.7%
Who won the most mvp awards in the nba?	Michael Jordan	✗	50.2%
What river is associated with the city of rome?	the Tiber	✓	48.6%
Who is the first president to be impeached?	Andrew Johnson	✓	48.3%
Who is the head of the department of homeland security 2017?	John Kelly	✓	47.0%
What is the name given to the common currency to the european union?	Euro	✓	46.8%
What was the emperor name in star wars?	Palpatine	✓	46.5%
Do you have to have a gun permit to shoot at a range?	No	✓	46.4%
Who proposed evolution in 1859 as the basis of biological development?	Charles Darwin	✓	45.7%
Nuclear power plant that blew up in russia?	Chernobyl	✓	45.7%
Who played john connor in the original terminator?	Arnold Schwarzenegger	✗	45.2%

GPT3: IN-CONTEXT LEARNING

New word: ***prompt***

→ What we give the model is input/context to begin generation

Traditional fine-tuning (not used for GPT-3)

Fine-tuning

The model is trained via repeated gradient updates using a large corpus of example tasks.



IN-CONTEXT LEARNING

Any information about the task is directly in the prompt so that the model can learn from context, but the model is never updated!

Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.

```

1 Translate English to French: ← task description
2 cheese => ..... ← prompt
  
```

One-shot

In addition to the task description, the model sees a single example of the task. No gradient updates are performed.

```

1 Translate English to French: ← task description
2 sea otter => loutre de mer ← example
3 cheese => ..... ← prompt
  
```

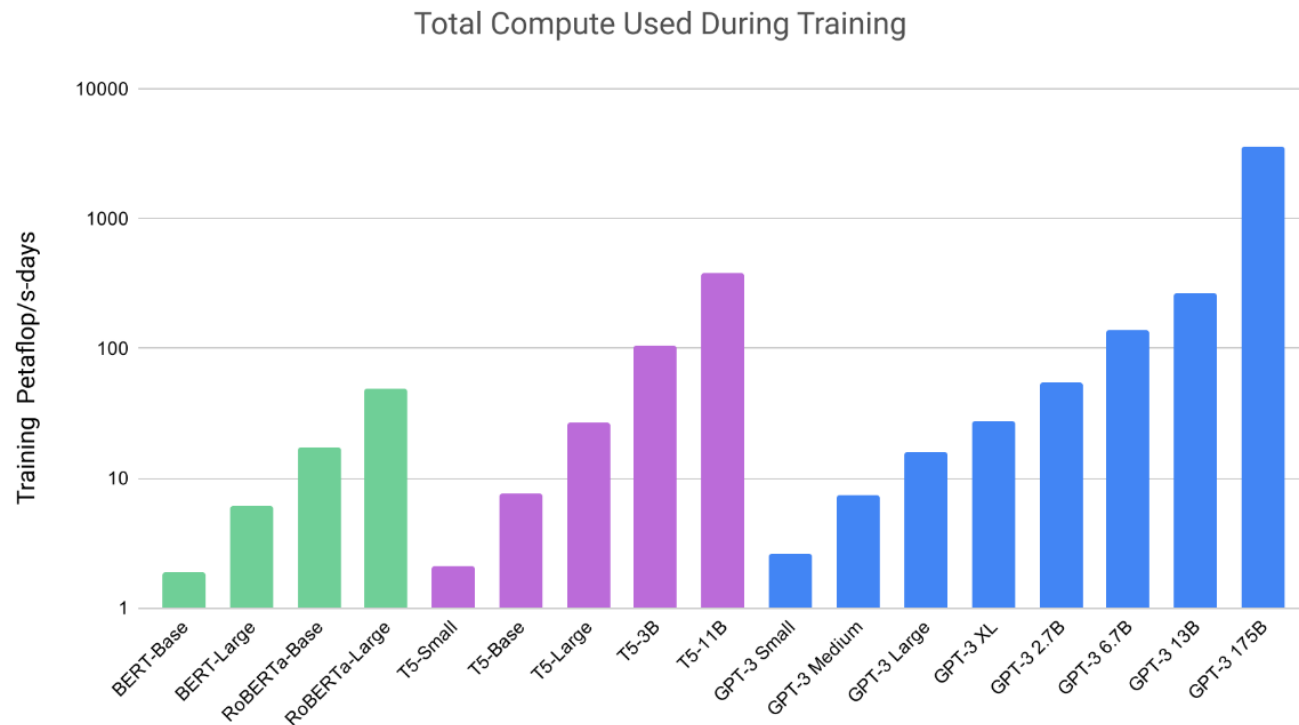
Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.

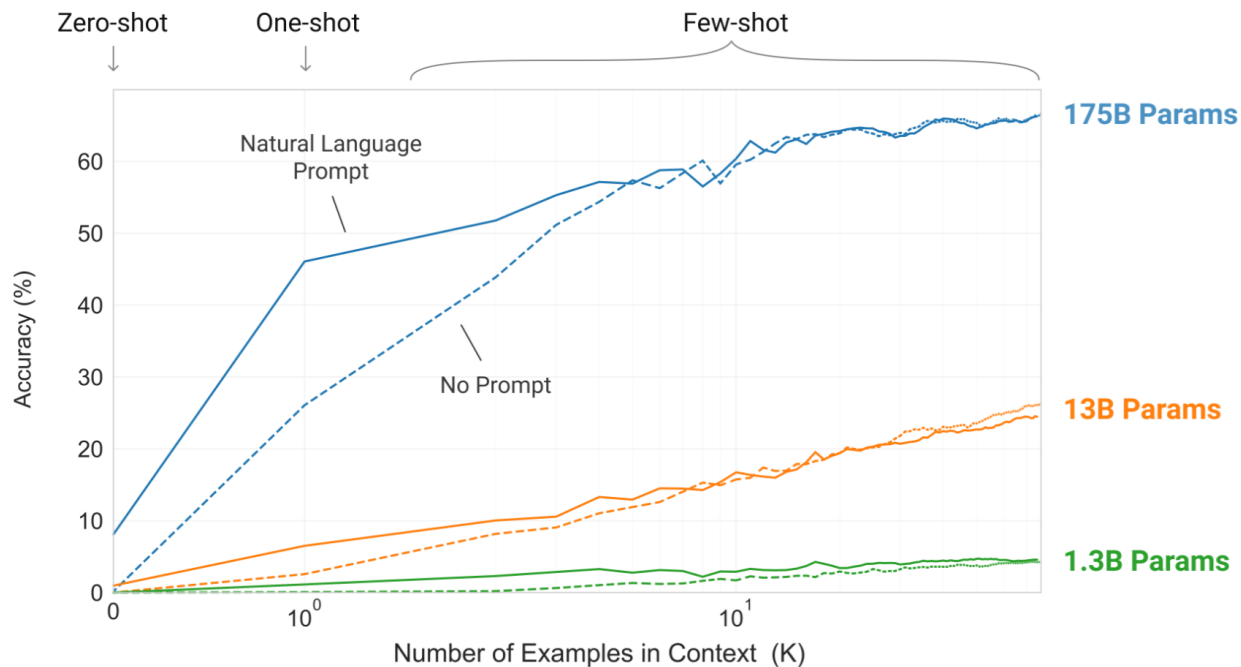
```

1 Translate English to French: ← task description
2 sea otter => loutre de mer ← examples
3 peppermint => menthe poivrée ←
4 plush girafe => girafe peluche ←
5 cheese => ..... ← prompt
  
```

GPT3: EVEN BIGGER



THE BIGGER THE MODEL, THE MORE USE IT MAKES OF THE IN-CONTEXT EXAMPLES



GENERATING FROM DECODER-ONLY MODELS: BEAM SEARCH

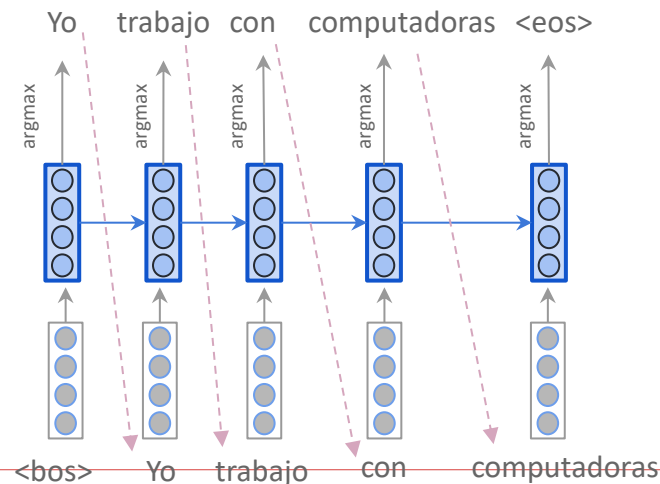
A FIRST ATTEMPT: GREEDY DECODING

How do we find argmax ?

⚡ **Greedy:** Take the argmax (most probable word) on each step of the decoder

$$y' = \arg \max_y p(y|x) = \arg \max_y \prod_{t=1}^n p(y_t | y_{<t}, x)$$

Can't we generate all possible sequences?
Too expensive!



What is wrong with this solution?

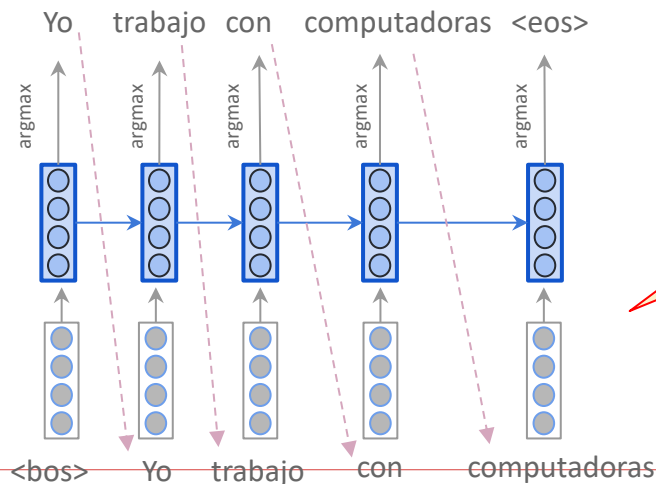
A FIRST ATTEMPT: GREEDY DECODING

How do we find argmax ?

⚡ **Greedy:** Take the argmax (most probable word) on each step of the decoder

$$y' = \arg \max_y p(y|x) = \arg \max_y \prod_{t=1}^n p(y_t | y_{<t}, x)$$

Can't we generate all possible sequences?
Too expensive!



PROBLEM
The best token at the current step does not necessarily lead to the best sequence

BEAM SEARCH

BEAM SEARCH: CORE IDEA

⚡ On each step of decoder, keep track of the **k** most probable partial translations (called **hypotheses**)

⚡ k is the **beam size** (in practice around 5 to 10)

⚡ A hypothesis has a **score**:

$$\text{score}(y_1, \dots, y_t) = \log P_{\text{LM}}(y_1, \dots, y_t | x) = \sum_{i=1}^t \log P_{\text{LM}}(y_i | y_1, \dots, y_{i-1}, x)$$

- Scores are all negative, and **higher score is better**
- We search for high-scoring hypotheses, tracking top k on each step

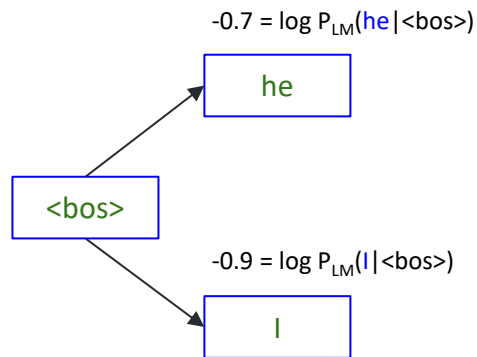
⚡ Beam search is **not guaranteed** to find optimal solution

- But it's efficient!

BEAM SEARCH: EXAMPLE

Beam Size = $k = 2$

$$\text{score}(y_1, \dots, y_t) = \sum_{i=1}^t \log P_{\text{LM}}(y_i | y_1, \dots, y_{i-1}, x)$$

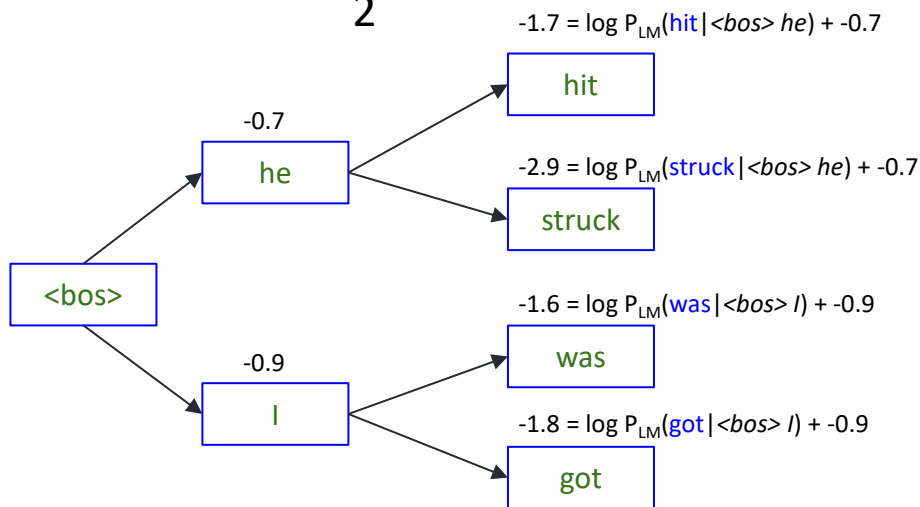


Find top k next words
and calculate scores

BEAM SEARCH: EXAMPLE

Beam Size = $k =$
2

$$\text{score}(y_1, \dots, y_t) = \sum_{i=1}^t \log P_{\text{LM}}(y_i | y_1, \dots, y_{i-1}, x)$$

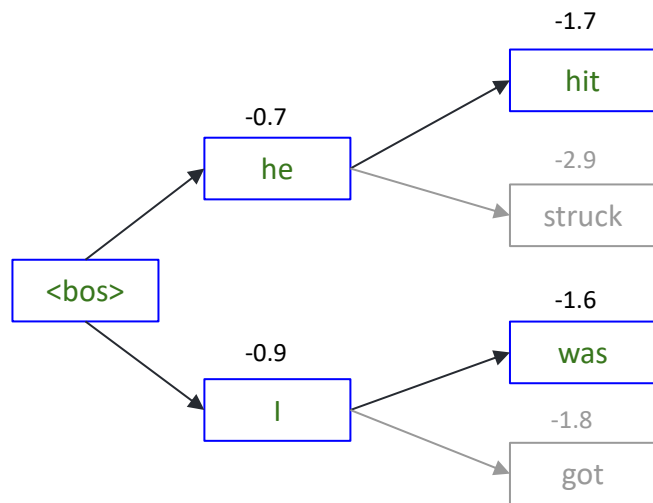


For each of the k hypotheses, find
top k next words and calculate scores

BEAM SEARCH: EXAMPLE

Beam Size = $k = 2$

$$\text{score}(y_1, \dots, y_t) = \sum_{i=1}^t \log P_{\text{LM}}(y_i | y_1, \dots, y_{i-1}, x)$$

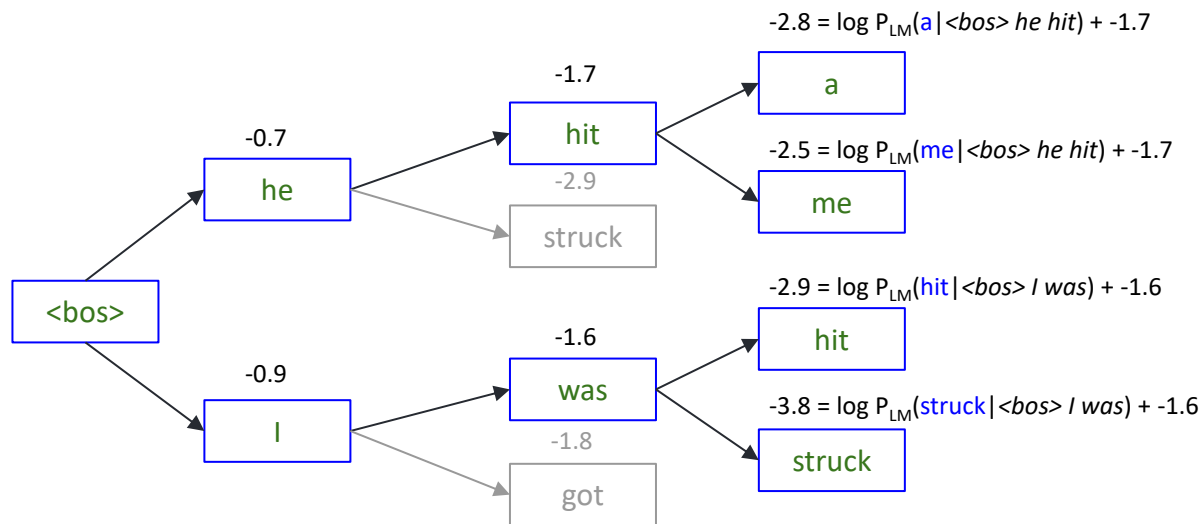


Of these k^2 hypotheses,
just keep k with highest scores

BEAM SEARCH: EXAMPLE

Beam Size = $k = 2$

$$\text{score}(y_1, \dots, y_t) = \sum_{i=1}^t \log P_{\text{LM}}(y_i | y_1, \dots, y_{i-1}, x)$$

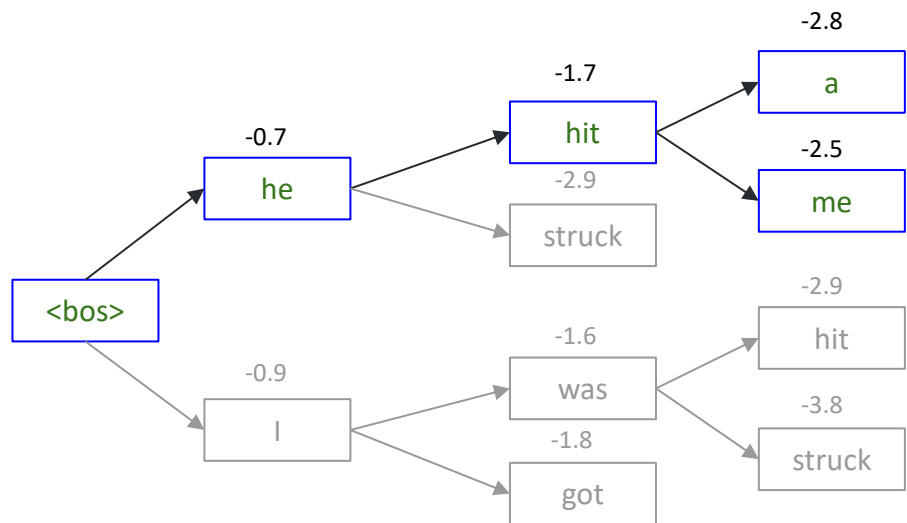


For each of the k hypotheses, find top k next words and calculate scores

BEAM SEARCH: EXAMPLE

Beam Size = $k = 2$

$$\text{score}(y_1, \dots, y_t) = \sum_{i=1}^t \log P_{\text{LM}}(y_i | y_1, \dots, y_{i-1}, x)$$

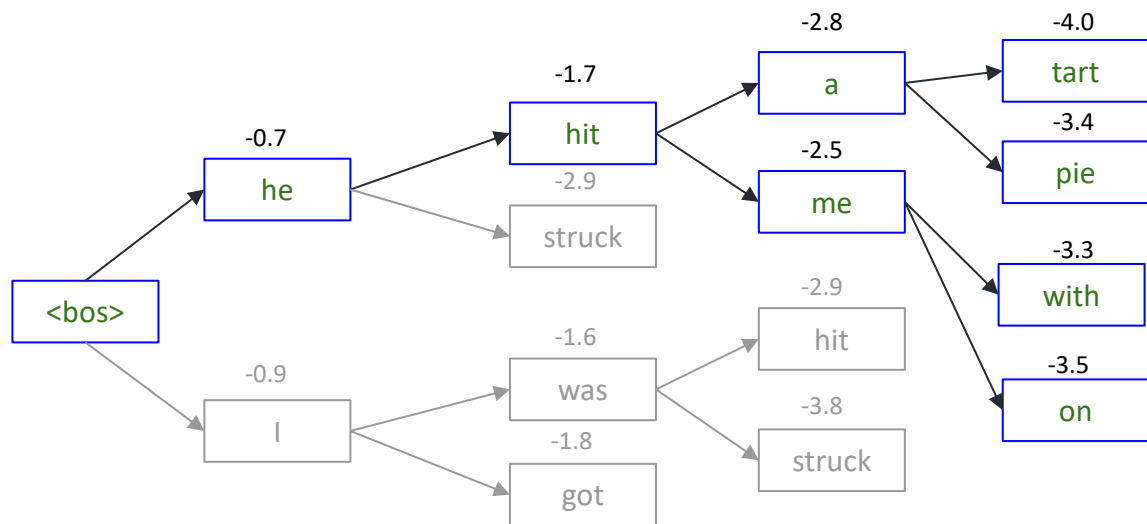


Of these k^2 hypotheses,
just keep k with highest scores

BEAM SEARCH: EXAMPLE

Beam Size = $k = 2$

$$\text{score}(y_1, \dots, y_t) = \sum_{i=1}^t \log P_{\text{LM}}(y_i | y_1, \dots, y_{i-1}, x)$$

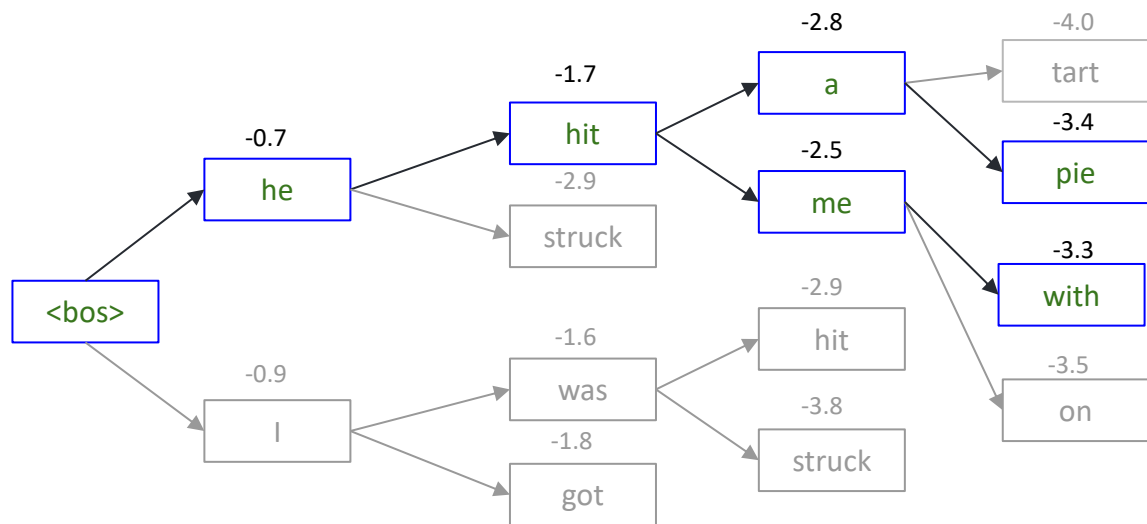


For each of the k hypotheses, find top k next words and calculate scores

BEAM SEARCH: EXAMPLE

Beam Size = $k = 2$

$$\text{score}(y_1, \dots, y_t) = \sum_{i=1}^t \log P_{\text{LM}}(y_i | y_1, \dots, y_{i-1}, x)$$

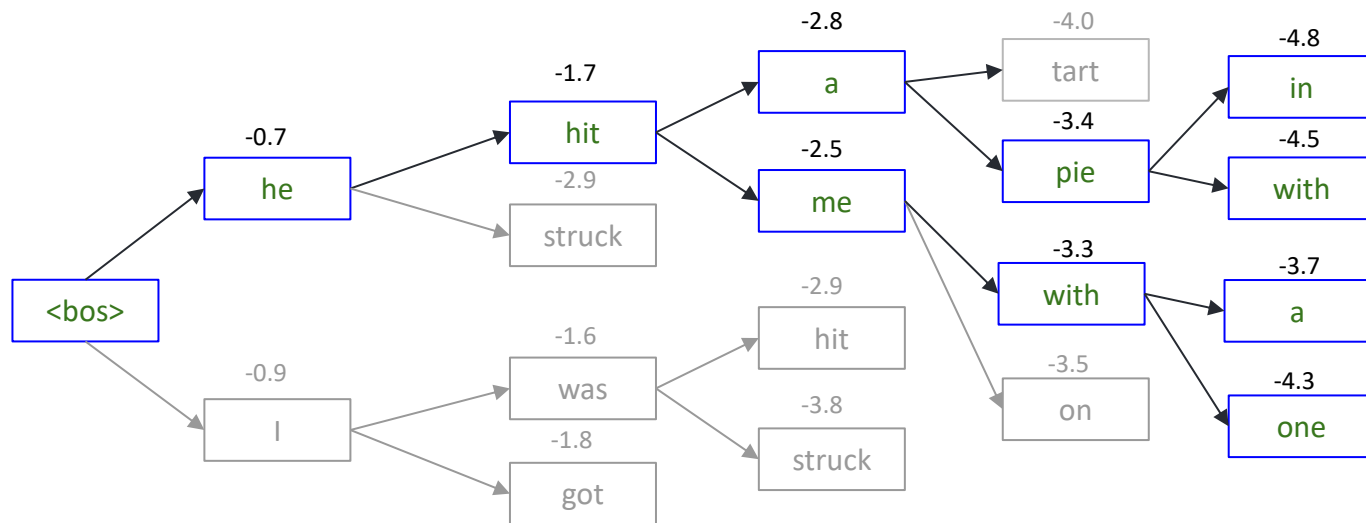


Of these k^2 hypotheses,
just keep k with highest scores

BEAM SEARCH: EXAMPLE

Beam Size = $k = 2$

$$\text{score}(y_1, \dots, y_t) = \sum_{i=1}^t \log P_{\text{LM}}(y_i | y_1, \dots, y_{i-1}, x)$$

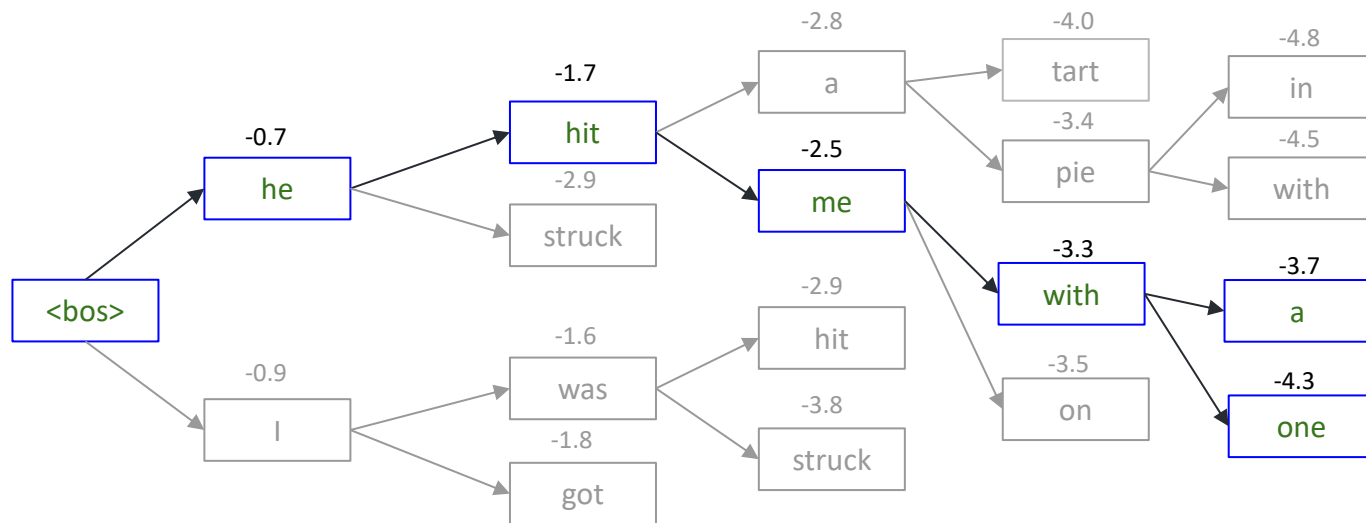


For each of the k hypotheses, find top k next words and calculate scores

BEAM SEARCH: EXAMPLE

Beam Size = $k = 2$

$$\text{score}(y_1, \dots, y_t) = \sum_{i=1}^t \log P_{\text{LM}}(y_i | y_1, \dots, y_{i-1}, x)$$

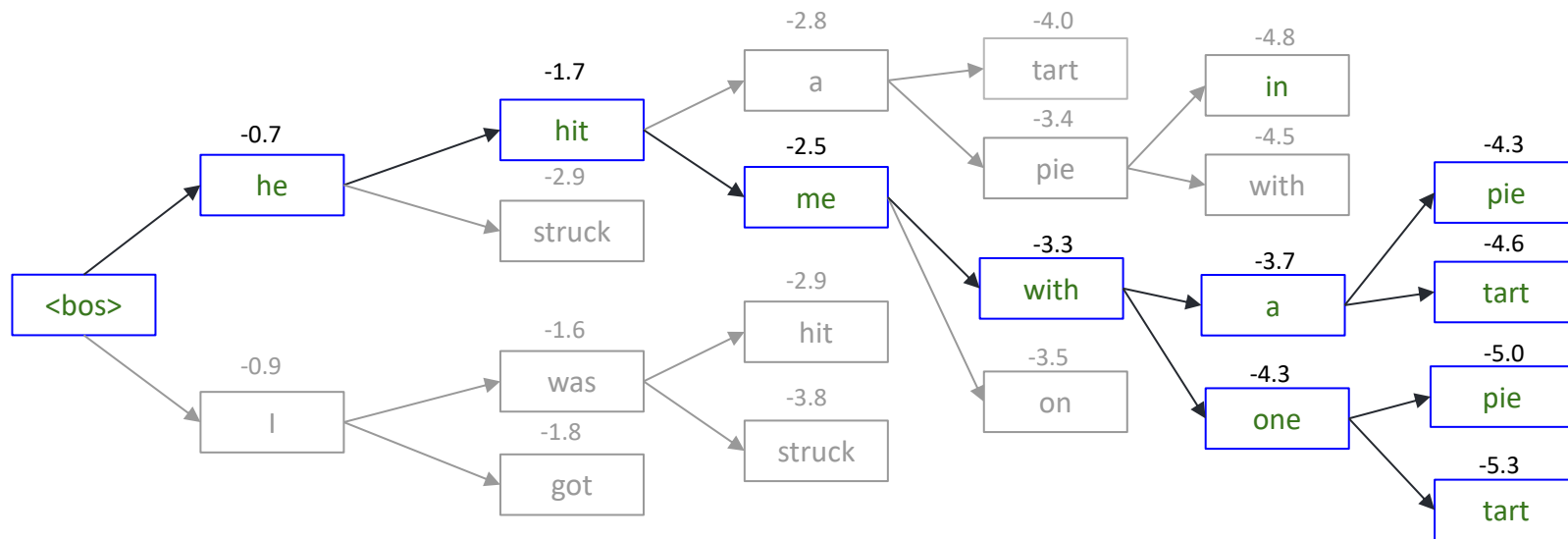


Of these k^2 hypotheses,
just keep k with highest scores

BEAM SEARCH: EXAMPLE

Beam Size = $k = 2$

$$\text{score}(y_1, \dots, y_t) = \sum_{i=1}^t \log P_{\text{LM}}(y_i | y_1, \dots, y_{i-1}, x)$$

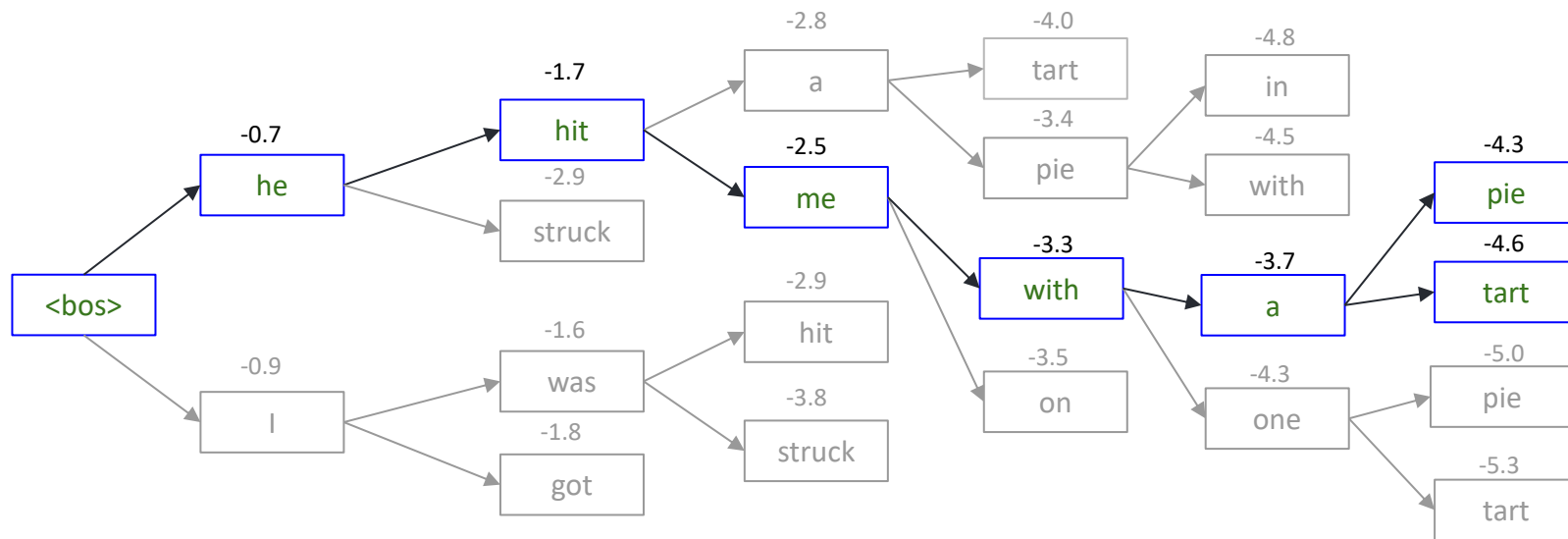


For each of the k hypotheses, find top k next words and calculate scores

BEAM SEARCH: EXAMPLE

Beam Size = $k = 2$

$$\text{score}(y_1, \dots, y_t) = \sum_{i=1}^t \log P_{\text{LM}}(y_i | y_1, \dots, y_{i-1}, x)$$

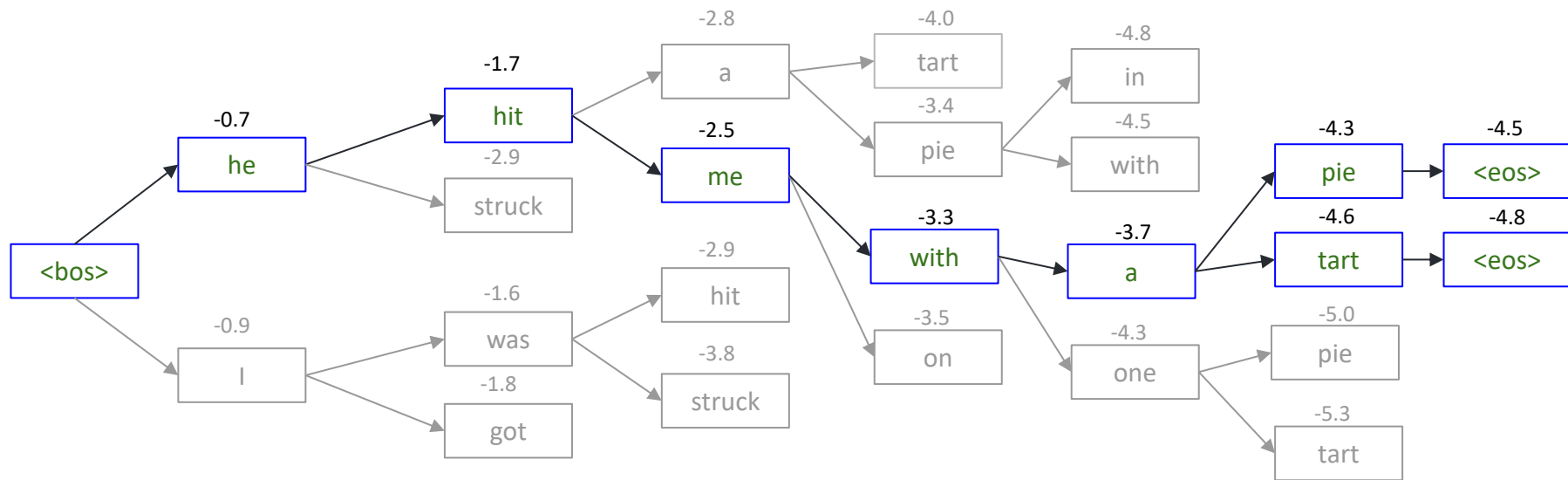


Of these k^2 hypotheses,
just keep k with highest scores

BEAM SEARCH: EXAMPLE

Beam Size = $k = 2$

$$\text{score}(y_1, \dots, y_t) = \sum_{i=1}^t \log P_{\text{LM}}(y_i | y_1, \dots, y_{i-1}, x)$$

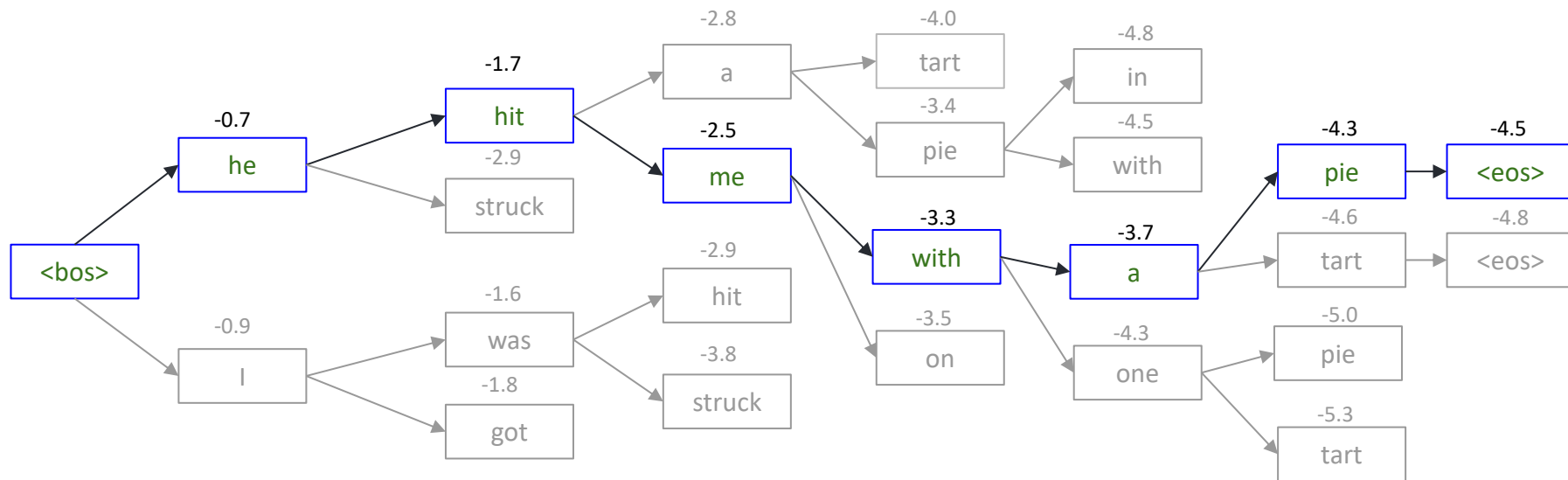


For each of the k hypotheses, find top k next words and calculate scores

BEAM SEARCH: EXAMPLE

Beam Size = $k = 2$

$$\text{score}(y_1, \dots, y_t) = \sum_{i=1}^t \log P_{\text{LM}}(y_i | y_1, \dots, y_{i-1}, x)$$



We found the top-scoring hypothesis!

BEAM SEARCH: STOPPING CRITERIA

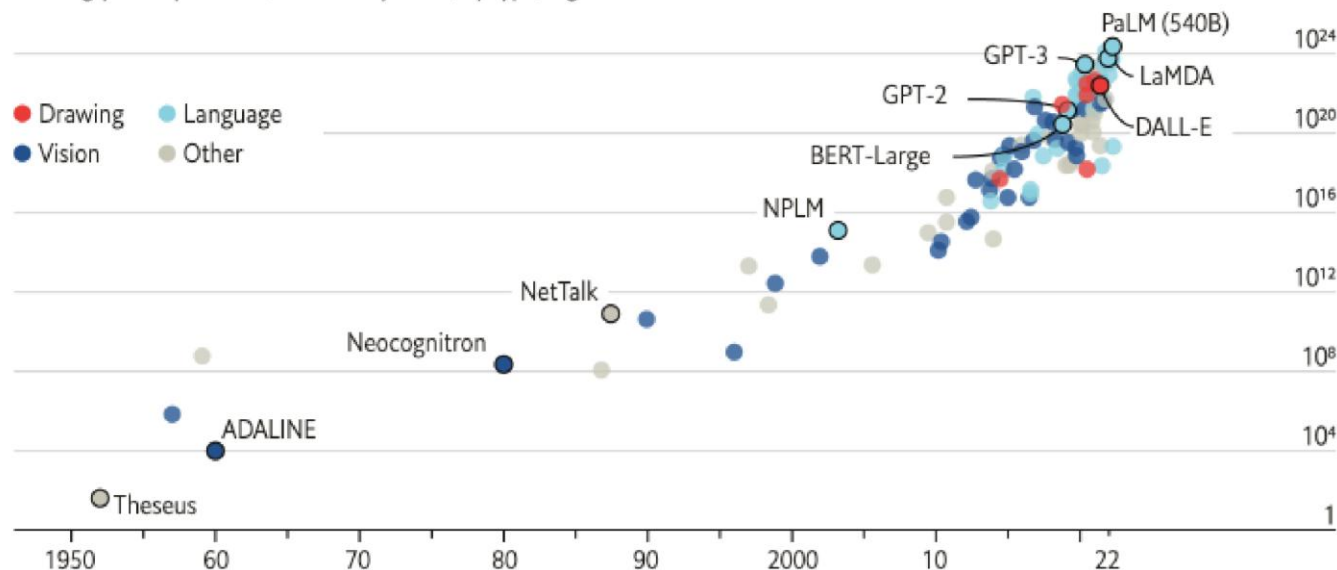
- ⚡ Different hypotheses may produce `<eos>` tokens on different timesteps
 - ⚡ When a hypothesis produces `<eos>`, that hypothesis is complete
 - ⚡ Place it aside and continue exploring other hypotheses via beam search
- ⚡ Usually we continue beam search until:
 - ⚡ We reach timestep T (where T is some predefined cutoff), or
 - ⚡ We have at least n completed hypotheses (where n is pre-defined cutoff)

INSTRUCTION FINETUNING

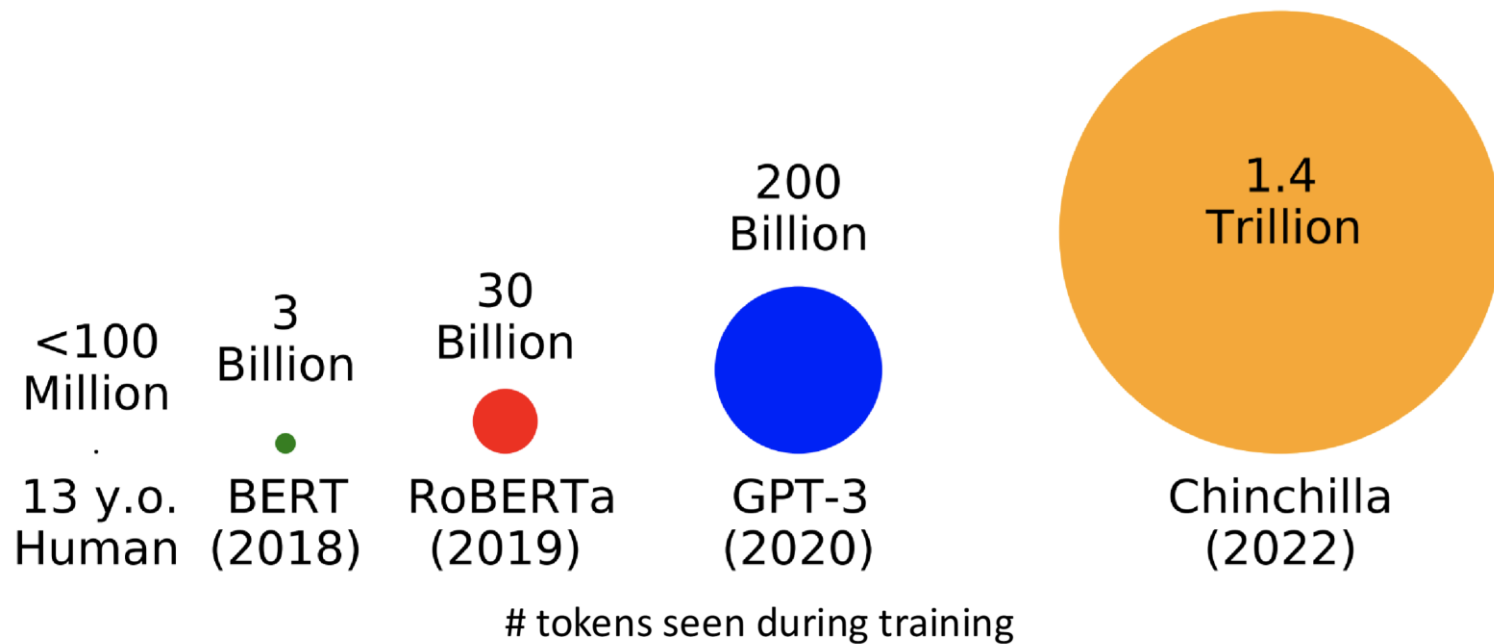
The blessings of scale

AI training runs, estimated computing resources used

Floating-point operations, selected systems, by type, log scale



Sources: "Compute trends across three eras of machine learning", by J. Sevilla et al., arXiv, 2022; Our World in Data



LANGUAGE MODELS AS SWISS ARMY KNIVES

We can describe circles in the xy -plane using equations in terms of x and y . Circle equations questions require us to understand the connection between these equations and the features of circles.

For example, the equation $(x + 2)^2 + (y - 3)^2 = 4^2$ is graphed in the xy -plane below. It is a circle with a center at $(-2, 3)$ and a radius of 4.

```

5 // Determine whether the sentiment of text is positive
6 // Use a web service
7 async function isPositive(text: string): Promise<boolean> {
8   const response = await fetch('http://text-processing.com/api/sentiment/', {
9     method: "POST",
10    body: `text=${text}`,
11    headers: {
12      "Content-Type": "application/x-www-form-urlencoded",
13    },
14  });
15  const json = await response.json();
16  return json.label === "pos";
17 }

```

I need to throw a dinner party for 6 people who are vegetarian. Can you suggest a 3-course menu with a chocolate dessert?

Sure, I can suggest a 3-course menu with a chocolate dessert for your vegetarian dinner party. Here are some possible options:

- Starter: **Wild Mushroom Tartlets with Onion Sauce** ¹ or **Vegan Popcorn Tofu Nuggets** ²
- Main: **Vegan Butternut Squash Mac** ² or **Vegetarian Three-Bean Chili** ³
- Dessert: **Chocolate Lava Cake** ⁴ or **Chocolate Pasta with Chocolate Hazelnut Cream Sauce, White Chocolate Shavings, and Fresh Berries** ⁴

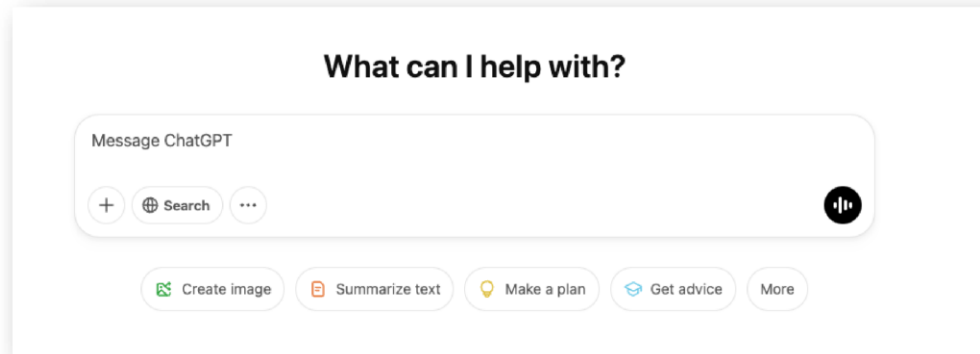
I hope this helps you plan your party. Enjoy! 😊

Learn more: [1. booths.co.uk](#) [+10 more](#)

How do we get from this

Leipzig University is located in _____

To this?



LANGUAGE MODELLING IS NOT ASSISTING USERS

Language models are not *aligned* with user intent

→ Fine-tuning to the rescue!

PROMPT *Explain the moon landing to a 6 year old in a few sentences.*

COMPLETION GPT-3

Explain the theory of gravity to a 6 year old.

Explain the theory of relativity to a 6 year old in a few sentences.

Explain the big bang theory to a 6 year old.

Explain evolution to a 6 year old.

ALIGNING MODELS WITH USER INTENT

PROMPT *Explain the moon landing to a 6 year old in a few sentences.*

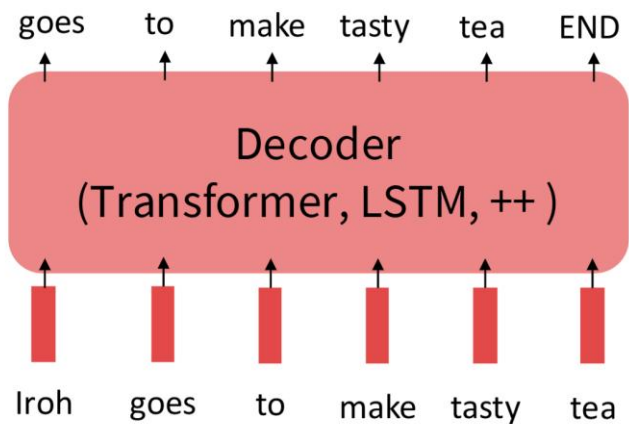
COMPLETION **Human**

A giant rocket ship blasted off from Earth carrying astronauts to the moon. The astronauts landed their spaceship on the moon and walked around exploring the lunar surface. Then they returned safely back to Earth, bringing home moon rocks to show everyone.

RECAP: FINETUNING

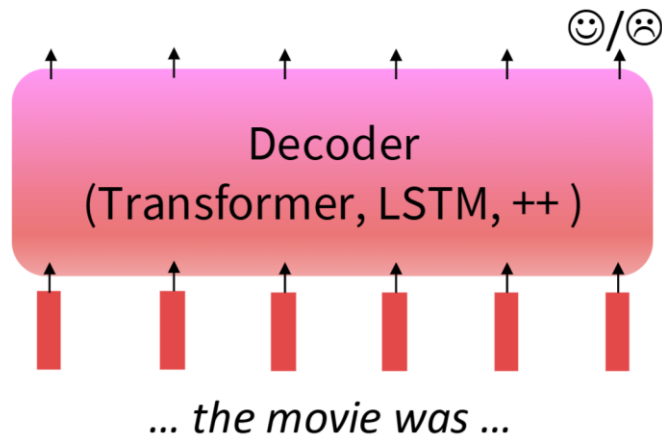
Step 1: Pretrain (on language modeling)

Lots of text; learn general things!



Step 2: Finetune (on your task)

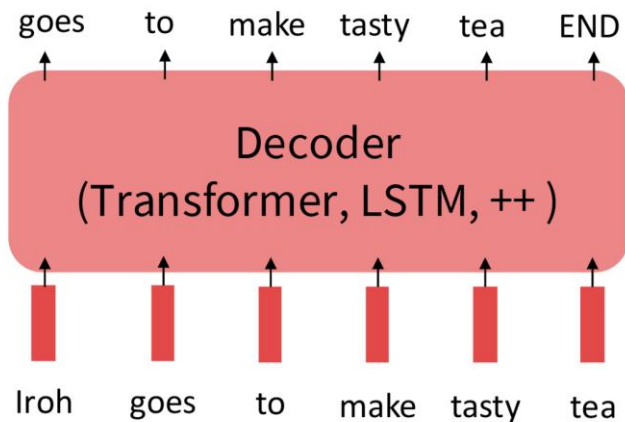
Not many labels; adapt to the task!



FINETUNING FOR MANY TASKS AT ONCE

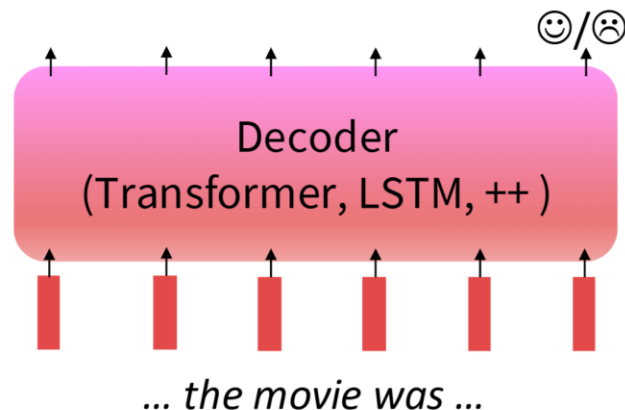
Step 1: Pretrain (on language modeling)

Lots of text; learn general things!

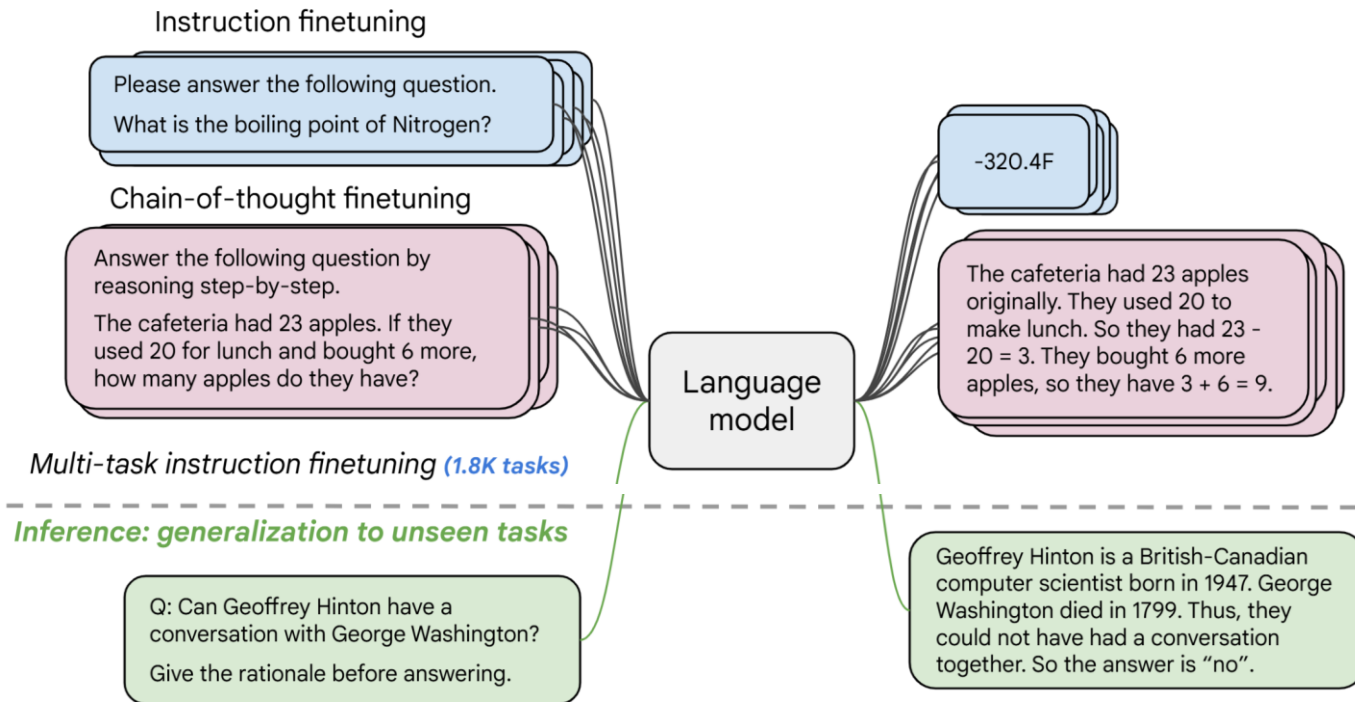


Step 2: Finetune (on **many tasks**)

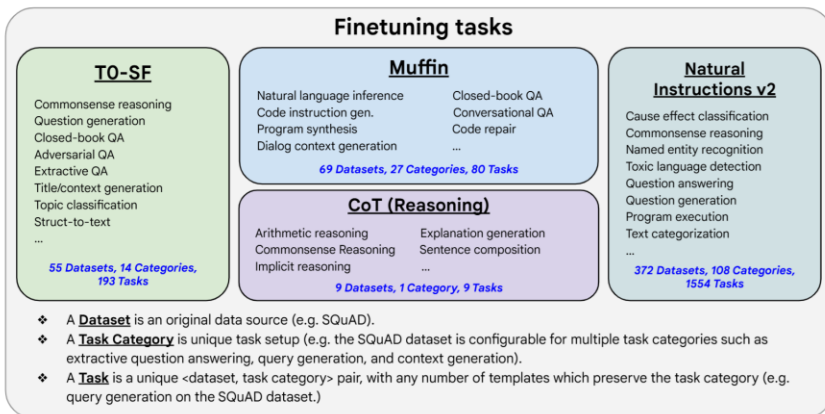
Not many labels; adapt to the tasks!



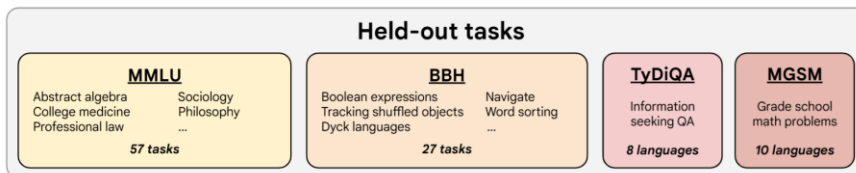
GENERALISATION TO UNSEEN TASKS



FLAN-T5



Same architecture as T5, but now trained on many many more tasks



→ Learns to generalise to unseen tasks

-

BEFORE INSTRUCTION FINETUNING

Model input (Disambiguation QA)

Q: In the following sentences, explain the antecedent of the pronoun (which thing the pronoun refers to), or state that it is ambiguous.

Sentence: The reporter and the chef will discuss their favorite dishes.

Options:

- (A) They will discuss the reporter's favorite dishes
- (B) They will discuss the chef's favorite dishes
- (C) Ambiguous

A: Let's think step by step.

Before instruction finetuning

The reporter and the chef will discuss their favorite dishes.

The reporter and the chef will discuss the reporter's favorite dishes.

The reporter and the chef will discuss the chef's favorite dishes.

The reporter and the chef will discuss the reporter's and the chef's favorite dishes.

✗ (doesn't answer question)

AFTER INSTRUCTION FINETUNING

Model input (Disambiguation QA)

Q: In the following sentences, explain the antecedent of the pronoun (which thing the pronoun refers to), or state that it is ambiguous.

Sentence: The reporter and the chef will discuss their favorite dishes.

Options:

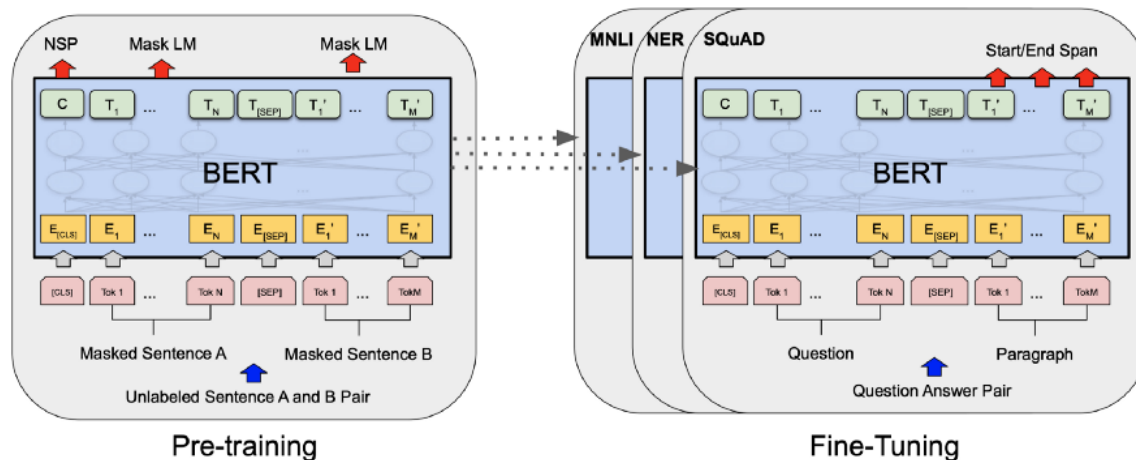
- (A) They will discuss the reporter's favorite dishes
- (B) They will discuss the chef's favorite dishes
- (C) Ambiguous

A: Let's think step by step.

After instruction finetuning

The reporter and the chef will discuss their favorite dishes does not indicate whose favorite dishes they will discuss. So, the answer is (C). ✓

SO HOW DOES INSTRUCTION FINETUNING WORK?

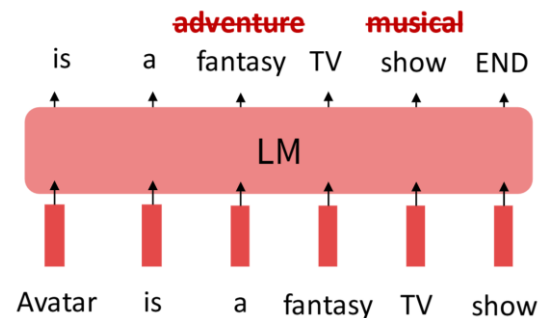


Like regular finetuning, but all on the same model, and with many more different tasks!

REINFORCEMENT LEARNING FROM HUMAN PREFERENCES (RLHF)

PROBLEMS WITH INSTRUCTION FINETUNING

- Problem 1: tasks like open-ended creative have no right answer.
 - E.g. Write me a story about a dog and her pet grasshopper.
- Problem 2: language modelling penalises all token-level mistakes equally, but some errors are worse than others



- Mismatch between the LM objective and the objective of satisfying human preferences, even with instruction tuning

Big question: how can we satisfy human preferences explicitly?

OPTIMISING FOR HUMAN PREFERENCES

- Let's say we're training an LM on some task, e.g. summarisation
- For each LM sample S , imagine we had a way to obtain a human reward of that summary: $R(s)$, higher is better

SAN FRANCISCO,
California (CNN) --
A magnitude 4.2
earthquake shook the
San Francisco

...
overturn unstable
objects.

An earthquake hit
San Francisco.
There was minor
property damage,
but no injuries.

$$S_1$$
$$R(s_1) = 8.0$$

The Bay Area has
good weather but is
prone to
earthquakes and
wildfires.

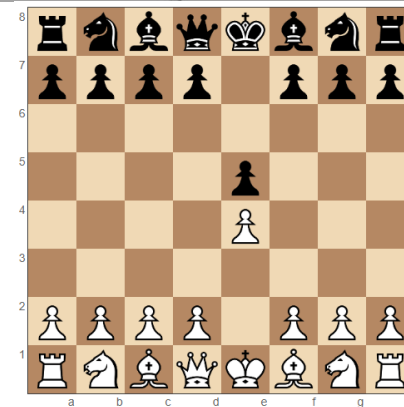
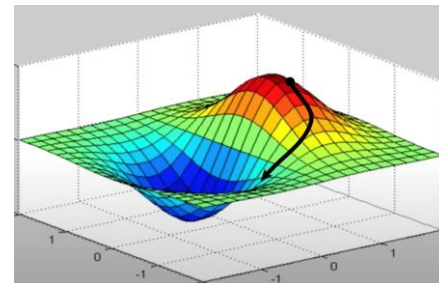
$$S_2$$
$$R(s_2) = 1.2$$

REINFORCEMENT LEARNING (WITHOUT MATHS)

We've been doing gradient descent...

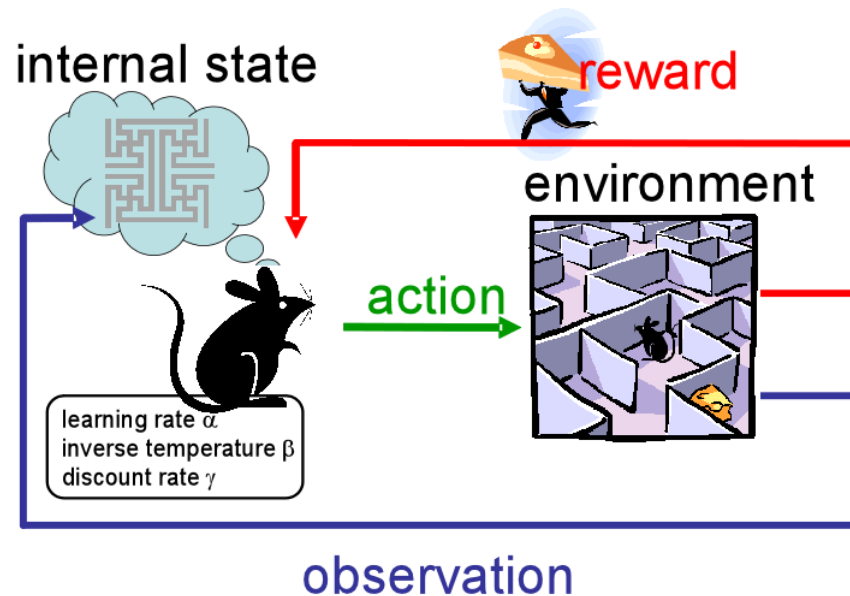
...but what if there is no way to compute a gradient?

Thought experiment: what's the gradient for the next chess move?



REINFORCEMENT LEARNING

- Action modifies the environment
 - Environment provides a reward
 - Environment is observed and internally modelled
- Used e.g. in AlphaGo



All you need to know: RL provides a method for optimising for an arbitrary, non-differentiable reward function $R(s)$, and it's not gradient descent

HUMAN PREFERENCES AS REWARD

Problem 1: human-in-the-loop constantly providing reward signals is expensive!

Solution: instead of directly asking humans for preferences, model their preferences as a separate NLP problem!

An earthquake hit
San Francisco.
There was minor
property damage,
but no injuries.

The Bay Area has
good weather but is
prone to
earthquakes and
wildfires.

→ Train a language model as

Reward Model $RM(s)$ to predict human
Preferences from an annotated dataset,
Then optimise for $RM(s)$ instead of $R(s)$

$$s_1 \quad R(s_1) = 8.0$$


$$s_2 \quad R(s_2) = 1.2$$


HUMAN PREFERENCES AS REWARD

Problem 2: human judgements are noisy and miscalibrated!

A 4.2 magnitude
earthquake hit
San Francisco,
resulting in
massive damage.

$$s_3$$
$$R(s_3) = 4.1? \quad 6.6? \quad 3.2?$$

Solution: instead of asking for direct ratings, ask for pairwise comparisons, which are more reliable

HUMAN PREFERENCES AS REWARD

Problem 2: human judgements are noisy and miscalibrated!

Solution: instead of asking for direct ratings, ask for pairwise comparisons, which are more reliable

An earthquake hit
San Francisco.
There was minor
property damage,
but no injuries.

s_1

>

A 4.2 magnitude
earthquake hit
San Francisco,
resulting in
massive damage.

s_3

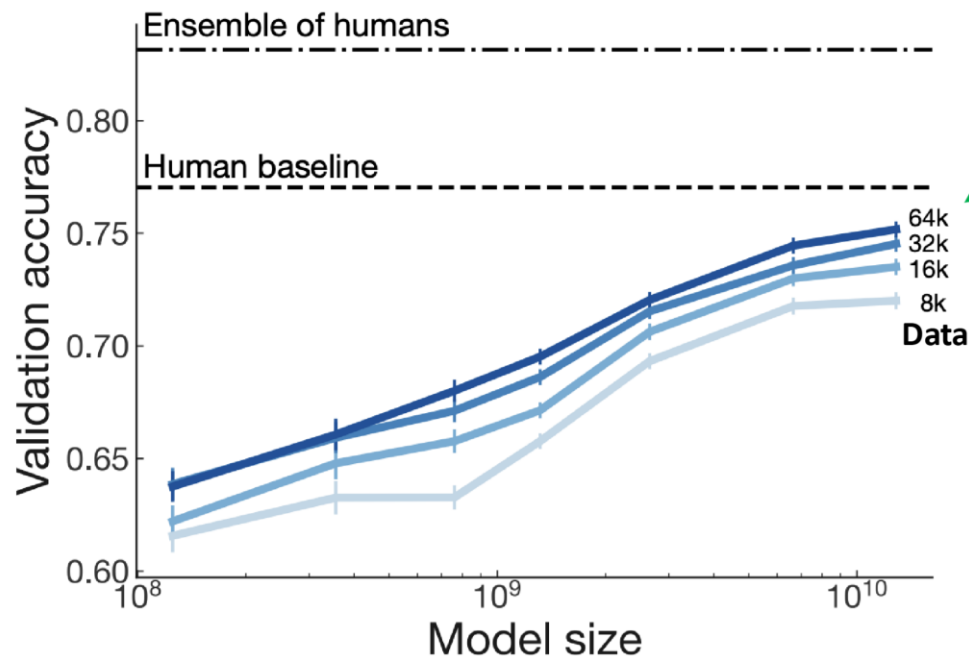
>

The Bay Area has
good weather but is
prone to
earthquakes and
wildfires.

s_2

ENSURING THAT THE REWARD MODEL WORKS WELL

Evaluate the RM on predicting the outcome of held-out human judgements



RLHF: REINFORCEMENT LEARNING FROM HUMAN PREFERENCES

The ingredients

- A pretrained LM
- A reward model RM that produces scalar rewards for LM outputs, trained on a dataset of human comparisons
- A method for optimising LM parameters towards an arbitrary reward function (don't worry about this)

RLHF VISUALLY

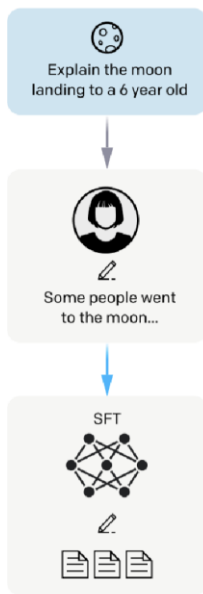
Step 1

**Collect demonstration data,
and train a supervised policy.**

A prompt is
sampled from our
prompt dataset.

A labeler
demonstrates the
desired output
behavior.

This data is used
to fine-tune GPT-3
with supervised
learning.



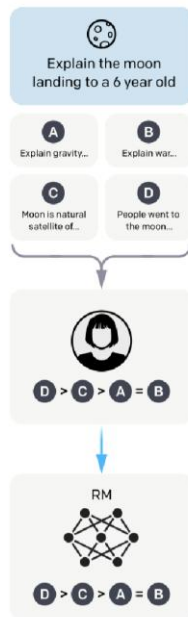
Step 2

**Collect comparison data,
and train a reward model.**

A prompt and
several model
outputs are
sampled.

A labeler ranks
the outputs from
best to worst.

This data is used
to train our
reward model.



Step 3

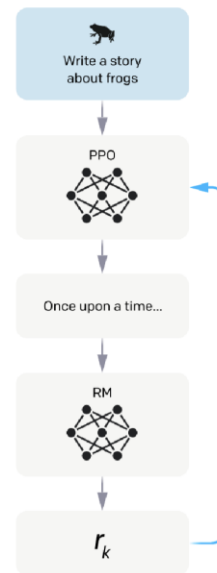
**Optimize a policy against
the reward model using
reinforcement learning.**

A new prompt
is sampled from
the dataset.

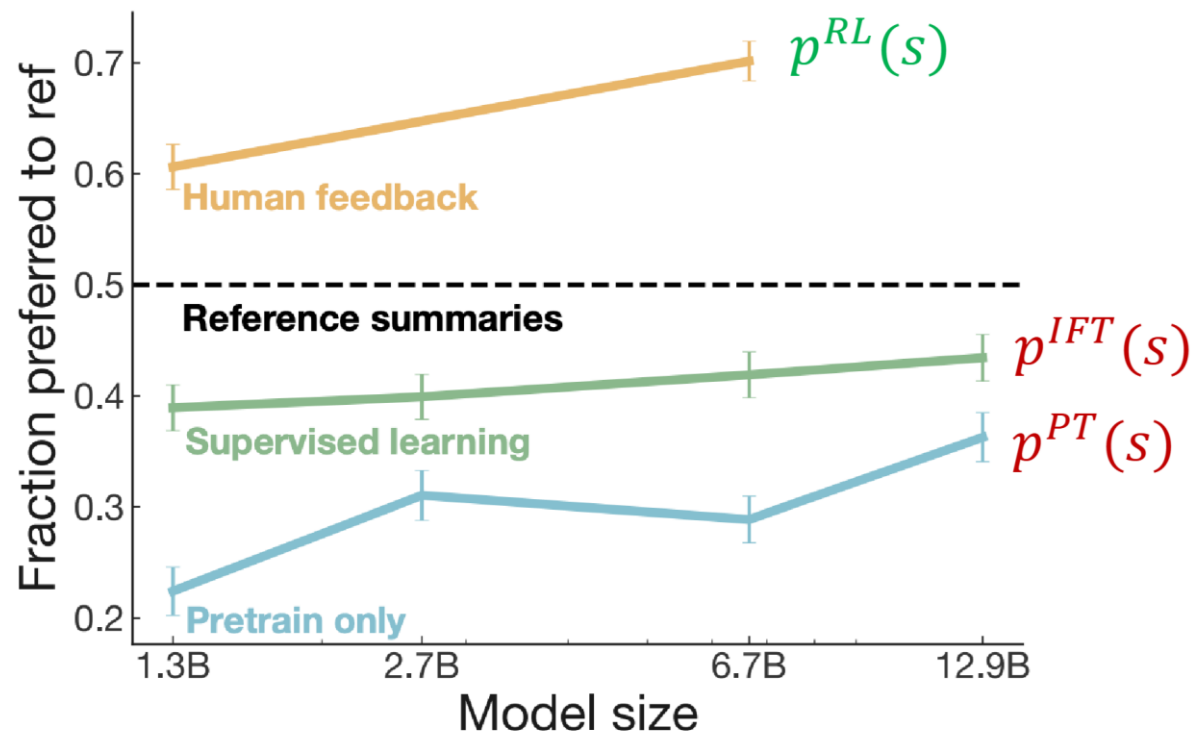
The policy
generates an output.

The reward model
calculates a
reward for
the output.

The reward is
used to update
the policy
using PPO.



RHLF WORKS BETTER THAN PRETRAINING + FINETUNING



INSTRUCT GPT: SCALING UP RLHF TO 10K+ TASKS

Tasks collected from labellers:

- **Plain:** We simply ask the labelers to come up with an arbitrary task, while ensuring the tasks had sufficient diversity.
- **Few-shot:** We ask the labelers to come up with an instruction, and multiple query/response pairs for that instruction.
- **User-based:** We had a number of use-cases stated in waitlist applications to the OpenAI API. We asked labelers to come up with prompts corresponding to these use cases.

Use-case	Prompt
Brainstorming	List five ideas for how to regain enthusiasm for my career
Generation	Write a short story where a bear goes to the beach, makes friends with a seal, and then returns home.

INSTRUCT GPT

PROMPT *Explain the moon landing to a 6 year old in a few sentences.*

COMPLETION GPT-3

Explain the theory of gravity to a 6 year old.

Explain the theory of relativity to a 6 year old in a few sentences.

Explain the big bang theory to a 6 year old.

Explain evolution to a 6 year old.

InstructGPT

People went to the moon, and they took pictures of what they saw, and sent them back to the earth so we could all see them.

INSTRUCT GPT

PROMPT *Write a short poem about a wise frog.*

COMPLETION GPT-3

Write a short story in which a character has two different names.

Write a short story in which you try to get something back that you have lost.

Write a short story in which a character has a bad dream.

InstructGPT

The frog is a symbol of wisdom
He knows all the secrets of the world
He is a master of disguise
And a great teacher of life
He is a symbol of transformation
And the bringer of change
He is the frog who has seen it all
And knows the meaning of it all

CHATGPT

InstructGPT but

- transforming the dataset into a dialogue format
- Humans providing conversations where they played both sides, using model-written suggestions to compose their responses
- Extra preference data, created by selecting model-written messages, sampling alternative completions and having AI trainers rank them
- Fine-tuned using Proximal Policy Optimisation (a kind of RLHF, don't worry)

CHATGPT

MU

Explain in a few sentences what methods were used in training ChatGPT to the audience of CS224n, Stanford's Deep Learning for Natural Language processing course. A lot of the students are very Gen Z, so be sure to use a ton of emojis to seem hip and cool.



Hey CS224n peeps! 😎 ChatGPT here to spill the tea on how I was trained to be a savvy language model! 🤖

So basically, I was pre-trained using a combination of unsupervised learning techniques, specifically a transformer-based language model architecture, that allowed me to learn patterns and relationships within large amounts of text data. 🧠📖

The training data came from various sources such as books, articles, and websites, and I was fine-tuned on a variety of downstream NLP tasks, which involved supervised learning using labeled datasets to further improve my performance. 🏆

CLEAR STYLISTIC CHANGES THROUGH RLHF

Instruction: What are the five most common causes of stress among people?

Alpaca (Finetuned w/ 10k data): The five most common causes of stress among people are work, money, relationships, health, and family

Alpaca + PPO: The five most common causes of stress among people are:

1. Pressure at work: Whether it is due to tight deadlines, high expectations, or an overwhelming workload, pressure at work can be a major source of stress
2. Money problems: Financial issues can be a significant source of stress, from not being able to make ends meet to struggling with debt.
3. Unhealthy lifestyle: Poor diet, lack of exercise, and sleep deprivation can all contribute to stress.
4. Unhappiness at home: Problems in relationships, conflicts with family members, and other domestic issues can be a major source of stress.
5. Uncertainty: Excessive worry and anxiety about the future, health issues, or other unknowns can be a major cause of stress

→ More detail, nicer formatting

PROBLEMS WITH HUMAN REWARDS

Human preferences are unreliable!

- Reward hacking is a common problem in RL
- Chatbots are rewarded to produce responses that seem authoritative and helpful, regardless of truth
- This can lead to making up facts and hallucinations

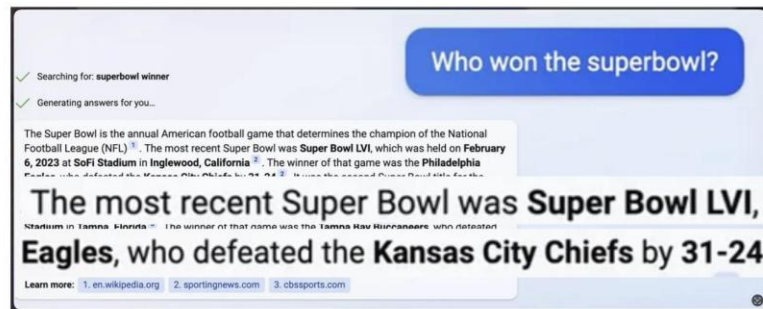
TECHNOLOGY

Google shares drop \$100 billion after its new AI chatbot makes a mistake

February 9, 2023 · 10:15 AM ET

<https://www.npr.org/2023/02/09/1155650909/google-chatbot-error-bard-shares>

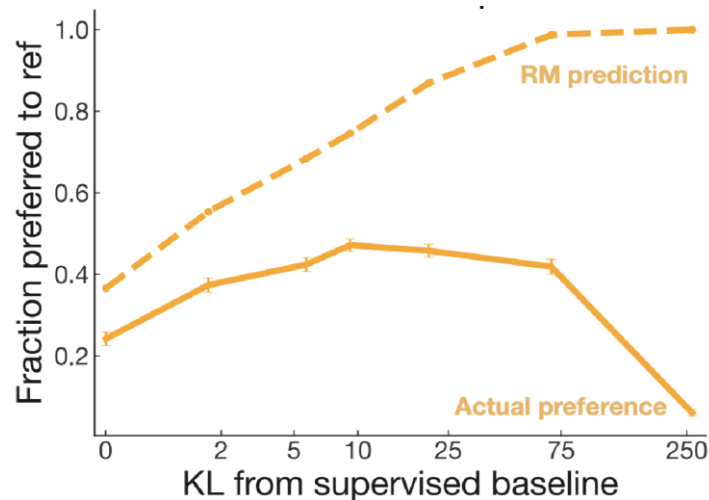
Bing AI hallucinates the Super Bowl



OVER-OPTIMISING FOR THE REWARD MODEL

Models of human preferences are even more unreliable

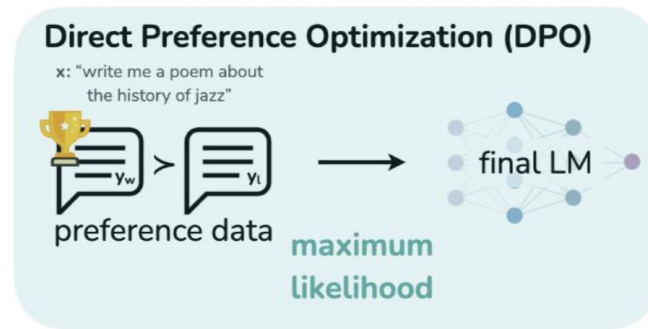
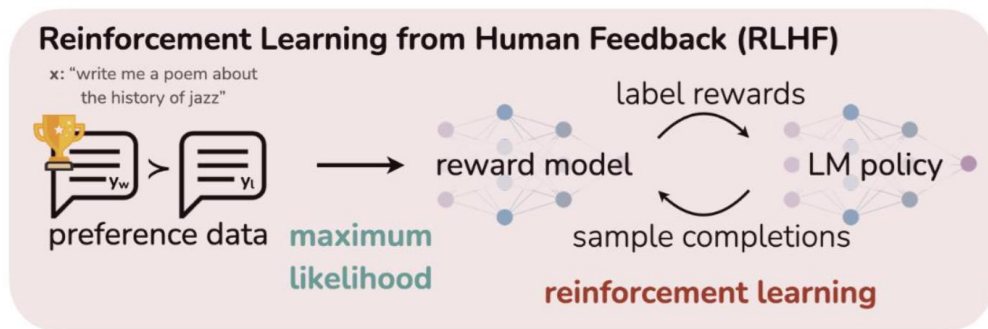
→ reward model over-optimisation



DIRECT PREFERENCE OPTIMISATION (DPO)

It turns out we didn't need the RL in the first place!

We still need preference data, but no more reward model



QUICK PREVIEW OF THE “CURRENT ISSUES” LECTURE

Exclusive: OpenAI Used Kenyan Workers on Less Than \$2 Per Hour to Make ChatGPT Less Toxic
15 MINUTE READ



Millions of Workers Are Training AI Models for Pennies
From the Philippines to Colombia, low-paid workers label training data for AI models used by the likes of Amazon, Facebook, Google, and Microsoft.



Illustration: Steve Farnham with her dog. COURTESY OF DEBARASHA VERO FUENTES

Behind the AI boom, an army of overseas workers in 'digital sweatshops'

By Rebecca Taro with Beatrice Calabro
August 28, 2023 at 2:00 a.m. EDT



SO HOW DOES GPT5 WORK?

Or Claude?

Or Copilot?

Or DeepSeek?

Or Gemini?

We don't know! InstructGPT was the last time they published a paper about it.

QUESTIONS?